

High-Tech 5-Axis Grinding for Premium TECINMAK Punches

Quality, Affordability, and Speed Combined

A Quick Look at the Manufacturing Processes for Premium TECINMAK Punches

1 — Planning and Design

Immediately after your order reaches TECINMAK your customer representative opens your order in the system. Then, engineers complete the product design and planning process on the same day.

2 — Material Preparation

Material for your order is immediately supplied from our raw material stocks, from our ready-made, heat-treated steel bars. One of the most important features of TECINMAK for stamping punches is its very strong raw material stocks and all steel bars are ready in our stocks in heat-treated form. After your order is processed by the raw material warehouse, this raw material is delivered to the production facility with a barcoded process document.

3 — Grinding & Machining Settings

The necessary adjustments are made by the operators on the 5-axis grinding machines. And production is started.

4 — Manufacturing Phase

Production is completed on our 5-AXIS CNC GRINDING machines with our special prescription and top-class raw materials.

5 — Quality Control and Inspections

After production is completed, all necessary checks are carried out by QC teams for each piece separately.

6 — TECINMAK's unique DRAG-FINISH process

For each piece - a free and special DRAG-FINISH process, which we code as G-Lap, is performed. TECINMAK always guarantees perfect surface quality. If your products have coating, this process is performed again after the coating phase.

7 — Delivery Phase

After G-Lap, a simplified final QC is done. The product is marked by the delivery team with laser marking. Then, each product is waxed to prevent damage during delivery. After labeling and packing, it is ready for delivery.

Ready Heat-Treated Steel Bars



Ready Solid Blanks



5-Axis Multi-grinding Machines & 6-Axis Wire EDM Machines of TECINMAK

















DEFENCE TOOLING for STAMPING

You can contact us for the most suitable solutions for your punches needs for the defence industry.
TECINMAK ***is an expert address in grinding the most complex geometries with up to 2 micron precision.***



"Turkish-Made / Globally Connected Quality"
We are proud to produce all of our products in our state-of-the-art center in Istanbul, Türkiye. Our facility is continuously updated with the latest advancements in manufacturing, inspection and material technology. The relationships we maintain with material, machine, and tool manufacturers worldwide enable us to stay at the forefront of industry technology ensuring that our facilities are among the most modern and well-equipped.

"The Perfect Combination"
We seamlessly integrate high-quality tool steel with advanced coatings to significantly extend tool service life. This optimal combination minimizes wear and material buildup, reducing your overall tooling costs. With this approach, your tools will deliver exceptional durability and performance, providing value and reliability that you've never experienced before."

ISO Certified:
To ensure the highest quality and meet customer requirements, we continuously improve our Quality Management System in accordance with ISO 9001:2015.









CUSTOM TOOLING and WIRE-EDM PARTS



CUSTOM CAN & END TOOLING SOLUTIONS FOR FOOD AND BEVERAGE



CARBIDE PUNCHES FOR BOLTS, NUTS AND PARTS



CUSTOM-MADE SELF-LUBRICATING COMPONENTS



We produce custom, precise, and in-house tooling parts for industries such as automotive, home appliances defence and aerospace, food and beverage, fasteners, door and window manufacturing, medical, electronic and many other sectors.

High-Performance Special Material Selection for Punches

Our core offering consists of these 4 steels, but we are also capable of providing the specific steel grade (or a close equivalent) you require, as well as Carbide. Unless otherwise specified, all punch and die buttons listed in our catalog and product code system will be manufactured from our M2 material.

Materials marked with * are available in our stock.

Steel Grade				Chemical Composition (%)								Feature (1 → 10 better)			
TECINMAK Steel Ref.	Related Standards			C	W	Mo	Mn	Si	Cr	V	Co	Wear Resistance	Hot Hardness	Toughness	Grindability
	AISI	VDEh	JIS												
*M2	M2	1.3343	SKH51	0.88	6.00	5.00	0.40	0.45	4.00	2.00	-	5	5	8	5
*PS (ASP30)	S590	1.3294	SKH40	1.30	6.00	5.00	0.50	0.70	4.00	3.00	8.00	7	9	9	9
*M42	M42	1.3247	SKH59	1.10	1.50	9.50	0.40	0.70	4.00	1.00	8.00	8	9	4	5
V4E	-	-	-	1.40	-	3.50	0.40	0.40	4.70	3.70	-	9	9	7	9

M2 — **M2** is a conventional high-speed steel, hardened and triple tempered to HRC:60-63. It has good toughness, wear resistance and high compression strength. Suitable for stamping thickness less than 5 mm.

PS (ASP30) — **ASP30** PM Steel is an extra-high-performance tool steel, designed for high-speed applications. Manufactured using powder metallurgy (PM) technology, this steel is one of the finest in its category, of ering exceptional properties such as high-temperature resistance, hardness, and wear resistance. This cobalt high-alloyed powder metallurgy steel offers excellent features for machining or cold work. Stable performance that certain grindability and less deformation after heat treatment. Excellent Hot Workability and becoming it is possible a high alloy. Superior wear resistance.

M42 — This material is especially preferred for use in screws, bolts, and fastener punches. **M42** is a high-speed steel alloy from the molybdenum series, containing an additional 8 to 10 percent cobalt. It is widely utilized in the metal manufacturing industry due to its superior red hardness compared to more conventional high-speed steels. This enhanced hardness allows for shorter cycle times in production environments, thanks to its higher cutting speeds and longer intervals between tool changes. **M42** is also less prone to chipping during interrupted cuts and is more cost-effective compared to carbide tools.

V4E — **Vanadis 4 Extra** Super Clean is one of the most versatile PM tool steels available in today's market, combining excellent wear resistance with high resistance to edge chipping and deformation. It offers outstanding machinability in both soft and hardened conditions, providing advantages in both tool manufacturing and tool maintenance.

Our Standard Surface Coatings for Ultimate Punch Life

"To receive the coating service, please add the 'Coating Code' to the end of the punch or order code"

TiN



- *TiN is the most cost-effective and widely used PVD coating.*
- *It offers suitable thermal stability for most cold working metal operations.*
- *TiN is ideal for light stamping processes with stamping lubricants but is not recommended for use with stainless steel, nickel, or copper applications.*

Coating Code: **Tin**

Material: Titanium nitride
Technology: PVD
Thickness Range: 1.3 µm - 3 µm
Colour: Gold

AlCrN Advanced Arc



- *Our most popular coating type.*
- *AlCrN offers excellent wear resistance and extended tool life.*
- *Ideal for stamping, piercing, blanking, and extrusions.*
- *Not recommended for aluminum.*
- *Significantly harder than traditional TiCN.*

Coating Code: **Alcrn**

Material: AlCrN
Technology: PVD Advanced Arc
Thickness Range: 2 - 4 µm
Colour: Grey

DUPLEX-VARIANTIC



- *Excellent results in cutting, forming, stamping and precision cutting.*
- *Low friction.*
- *High oxidation resistance.*
- *Extremely high wear resistance.*
- *Our top-performing premium coating option for durability and performance.*

Coating Code: **Duplex**

Material: TiAlCN
Technology: PVD Advanced Arc
Thickness Range: 4 - 6 µm
Colour: Old Rose

DLC

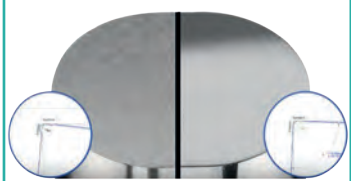


- *The DLC coating is excellent for stamping non-ferrous metals (aluminum etc.)*
- *Superior resistance to abrasive wear and galling.*
- *DLC performs best when combined with pre and final polishing.*

Coating Code: **Dlc**

Material: DLC
Technology: PACVD
Thickness Range: 2 - 4 µm
Colour: Dark-grey & Black

DRAG-FINISH (G-Lap)



For each piece - a free and special DRAG-FINISH process, which we code as G-Lap, is performed. TECINMAK ***always guarantees perfect surface quality. If your products have coating, G-Lap is performed again after the coating phase.***

TECINMAK **"G-Lap"** utilizes a specialized operation containing diamond particles to polish even the most irregular surfaces and hard-to-reach areas.

- Improves the durability of piercing and forming tools.
- Delivers a uniform surface finish without distortion or damage.
- Enhances the finish and increases adhesion for PVD/CVD coatings.
- Virtually no material is removed, ensuring tight tolerances are consistently maintained.

4 KEY BENEFITS OF CHOOSING STANDARD PUNCH AND MATRIX

In industrial metal forming operations, stamping components used for hole punching are often custom-designed for each product, leading to incompatibility with other press operations. Punch and matrix sets are typically one-off components designed specifically for each machine and operation. For instance, a set may be designed for a particular process. Designing custom punch and matrix sets for every product line is not an efficient approach for planning manufacturing equipment and capital. Instead, using standard punch and matrix sets for common manufacturing processes and workpiece materials is a much more efficient strategy. At this point, TECINMAK is an ideal partner for your projects, offering products compliant with DIN-ISO standards and fast delivery capabilities.

1. Improved Consistency and Dimensional Accuracy

Standard production tools provide better dimensional accuracy and precision compared to custom dies. Parts made with standardized die sets tend to have fewer defects, which stabilizes the production process.

2. Increased Cost Efficiency

Standard stamping components help reduce scrap and rework costs. Bulk ordering also results in significant savings on capital equipment. Storing a small number of components reduces inventory costs.

3. Simplified Maintenance and Replacement

Standard, interchangeable punch and matrix components simplify maintenance and replacement. When a part breaks, it can be quickly replaced with an existing spare part, eliminating the need for custom part fabrication.

4. Accelerated Design and Prototyping Processes

Standard components speed up the design and prototyping process. CAD models allow engineers to verify component fit early in the design process, reducing prototype production time and accelerating product development. These benefits make the production process much more efficient, resulting in significant cost savings and improved quality.

UNDERSTANDING DIE CLEARANCE FOR PUNCHES AND MATRIX

Punch and die clearance is the distance between the punch's cutting edge and the matrix cutting edge, critical for clean precise cuts in metal punching. This clearance ensures the punch effectively shears the material and creates a clean hole without excessive deformation or burrs.

"Importance of Punch and Die Clearance" Proper clearance ensures high-quality cuts with minimal burrs, reducing rework time and increasing production efficiency. It also minimizes tool wear, extending the lifespan of expensive tooling and reducing maintenance costs. Additionally, optimized clearance improves energy efficiency by requiring less force for cutting.

"Factors Affecting Clearance Selection"

Tooling Condition: The state of the tools impacts the clearance adjustment depending on tool wear.

Material: Harder materials require more clearance than softer ones.

Thickness: Thicker workpieces need more clearance.

Recommended Clearance for Common Materials:

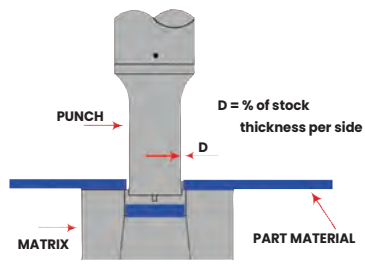
We suggest maintaining a cutting clearance between the matrix and the punch equal to 10% of the sheet thickness being processed. Stronger materials require higher clearances, while precise grades require lower clearances.

"Adjusting Clearance for Special Applications"

Fine Blanking: This technique demands very small clearances for high precision but results in higher tool wear.

Hard/High Tensile Materials: These require more clearance due to their strength, and tighter clearances increase tool wear over time.

In summary, punch and die clearance directly impacts cut quality, tool longevity, and production efficiency. Adjustments based on material and application type are crucial for optimal performance.



Recommended Clearance for Common Materials:

Material	Soft	Hard
Aluminum	% 10	% 12
Brass / Copper	% 6	% 8
Low Carbon Steel	% 10	% 12
High Carbon Steel	% 18	% 20

EXAMPLE: Recommended clearance 10% with 2,0 mm thick material (clearance = 0,20 mm):
Total punch-to-matrix clearance (= clearance per side * 2): 0,40 mm **Punch Point Size:** 12,60 mm & **Matrix Hole Size:** 13,00 mm

ABOUT EJECTOR PUNCHES

"TECINMAK Standard Solid Blank - Ejector Punch Series"

The **Ejector Punches** series, offered by TECINMAK as **standard solid blank,** ensures the **ejection of the slug (burr or chip) from the punch point** by integrating the ejector into the punch.

Applicable to **ISO 8020 series,** as well as **Ball Lock, Heavy-Duty Headed, Center-Dowel Headed, and Bottle-Neck** product groups.

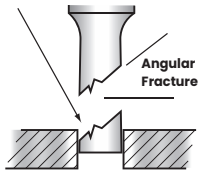
All models are in stock and ready for shipment.



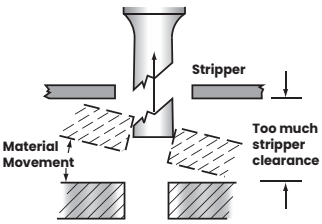
The system is composed of three main parts: **Ejector Pin, Spring, and Set Screw.** The spring on top of the ejector provides the necessary force for its function, while the set screw is tightened into the punch head to secure the entire assembly.
During operation, the punch first contacts the material. At the same time, the ejector pin is pushed back, compressing the spring. The punch then continues its normal function, creating the required hole in the material. Once the slug is separated, the spring expands, pushing the ejector pin outward and releasing the slug from the punch tip.

KEY FACTORS BEHIND PUNCH PROBLEMS

STRIPPING FAILURE



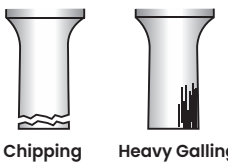
FAILURE:
A portion of the punch is broken off in the material.



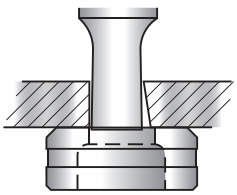
PROBABLE CAUSE:
Too much stripper clearance.

POSSIBLE SOLUTION: *Adjust stripper closer to the material.*

CHIPPING and GALLING



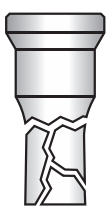
FAILURE:
Punch face chipping or heavy galling on one area of punch.



PROBABLE CAUSE:
Poor alignment between punch and matrix causing the punch to drag.

POSSIBLE SOLUTION: *Adjust alignment between punch and matrix to create equal clearance all around.*

COMPRESSIVE FAILURE

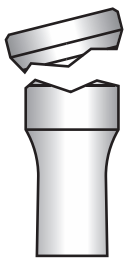


FAILURE: Occurs when the compressive strength of the punch has been exceeded and the entire working end fractures.

PROBABLE CAUSE:
Attempting to punch extremely hard or thick materials, or complete misalignment of the punch and die.

POSSIBLE SOLUTION: *Check the alignment, and if the issue is not with the alignment, consult* TECINMAK

PUNCH HEAD FAILURE

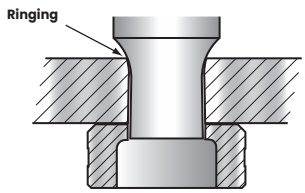


FAILURE: Punch head fractures or breaks off.

PROBABLE CAUSES and EXPLANATIONS:

- The material thickness should not be variable. The reason is that during the punch retraction process, if the punch remains stuck, sudden stress may cause the head to break off.
- For a smoother surface, the minimum step area of the punch should be coated. This smoothness allows the punch to move more easily.
- The head is very hard, and there is no tempering in it.
- Not using a matrix with the punch.
- Mounting the punch onto the plate instead of using a Punch Retainer.
- Using a loose or worn Punch Retainer.

RINGING



FAILURE: Material being punched is deformed with each stroke of the press.

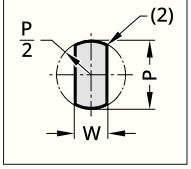
PROBABLE CAUSE:
Material is thicker than the working length of the punch, or the punch is entering into the matrix too far.

POSSIBLE SOLUTION: *Adjust the stroke length to enter into the matrix a maximum of 6 - 7%.*

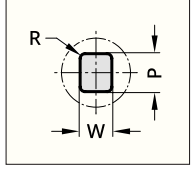
WARNING:
Ensuring the safe use of this tooling is the machine operator's responsibility.

Round, flattened

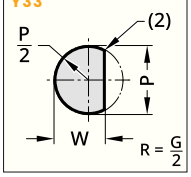
Y10



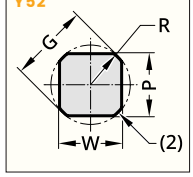
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Y33

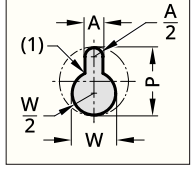


Y52

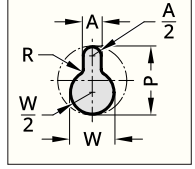


Key-hole shapes

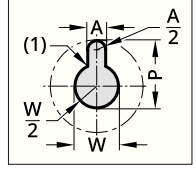
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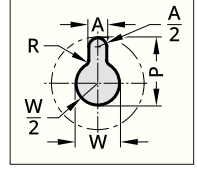
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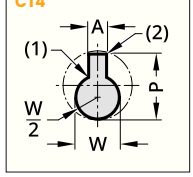
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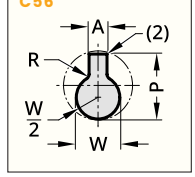
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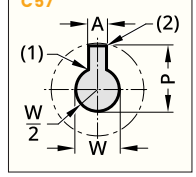
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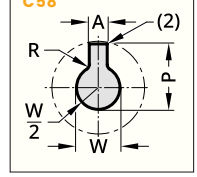
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C57

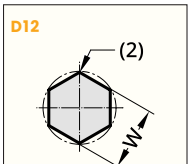


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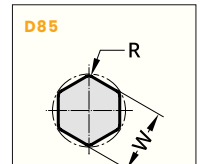


Polygons

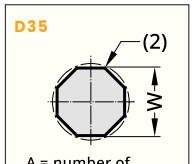
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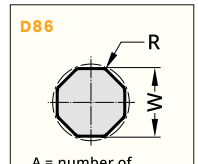
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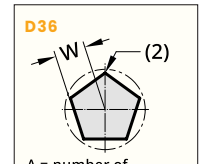
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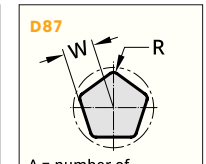
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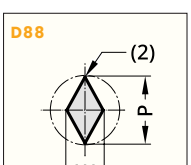
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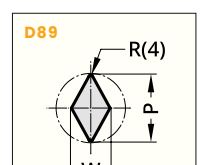
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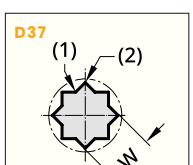
D88



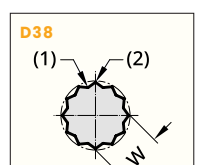
D89



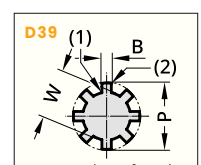
D37



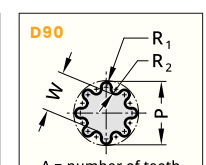
D38



D39

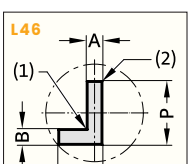


D90

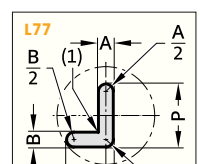


L-shapes

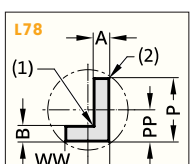
L46



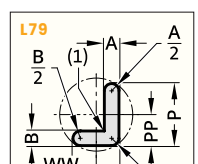
L77



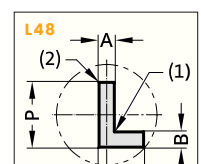
L78



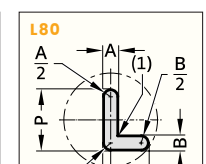
L79



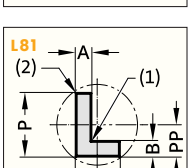
L48



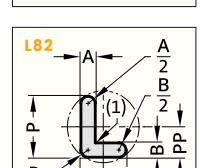
L80



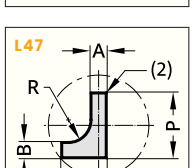
L81



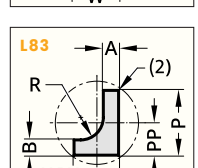
L82



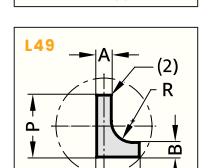
L47



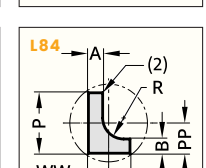
L83



L49

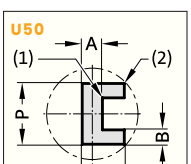


L84

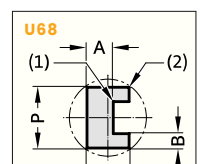


U-shapes

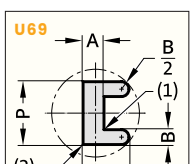
U50



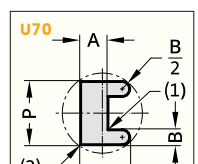
U68



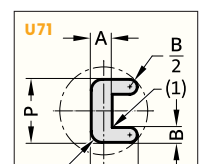
U69



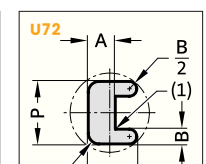
U70



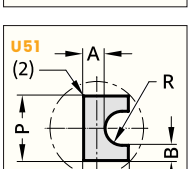
U71



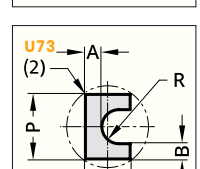
U72



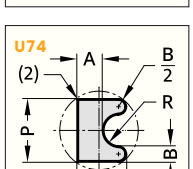
U51



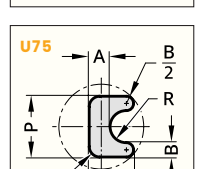
U73



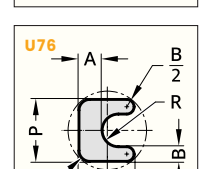
U74

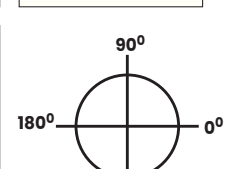


U75

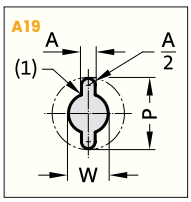


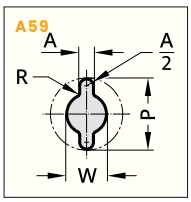
U76

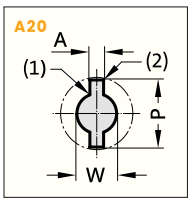


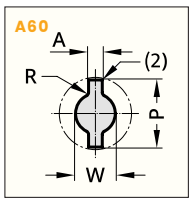


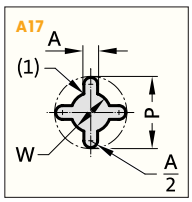
Multi key-hole shapes

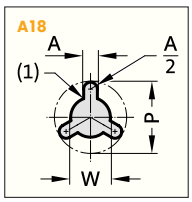




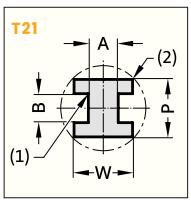


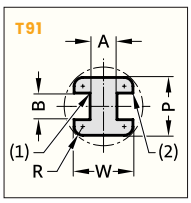


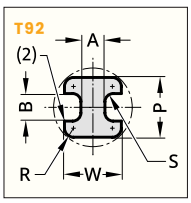


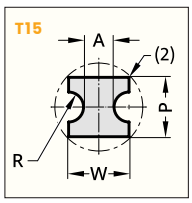


Double T-shapes

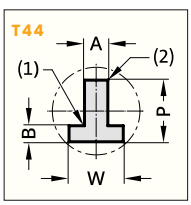


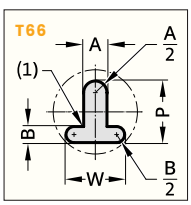


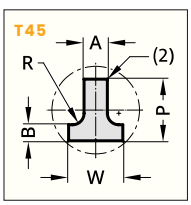


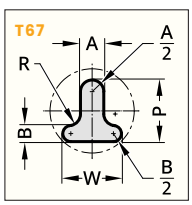


Simple T-shapes

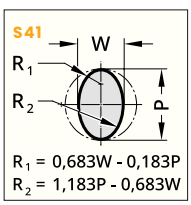


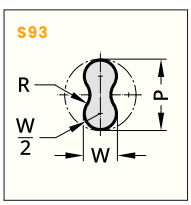


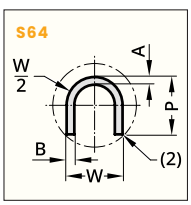


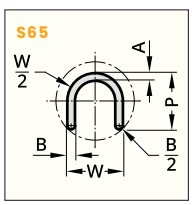


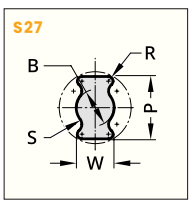
Various

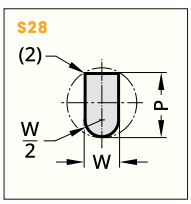


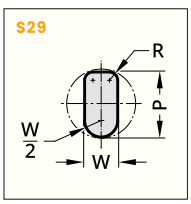


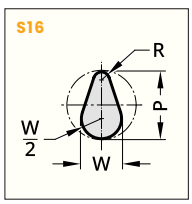


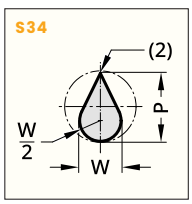




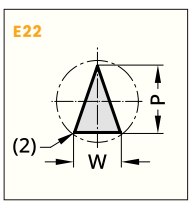


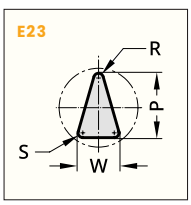


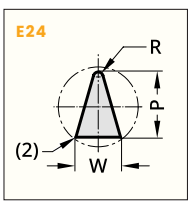


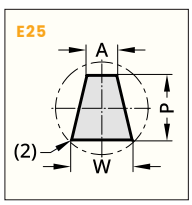


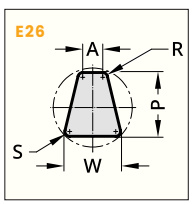
Triangles, trapezes



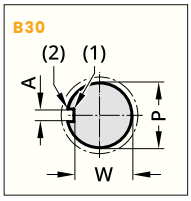


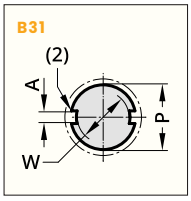


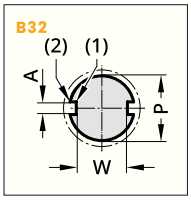


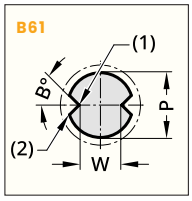


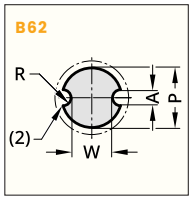
Key-hole



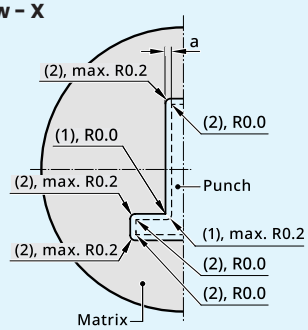






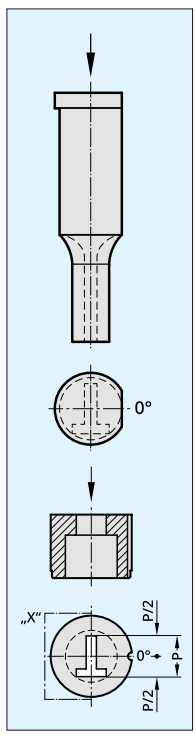


View - X



Cutting gap / Clearance (a)
Roundings with the corresponding sharp corners reduce the cutting gap per side **(a)**. If the cutting gap is 0.04 mm **(a)** or less, TECINMAK will round the sharp edges if the cutting punch and the matrixes are ordered together. This reduces the installation time and the risk of an edge breaking during operation.

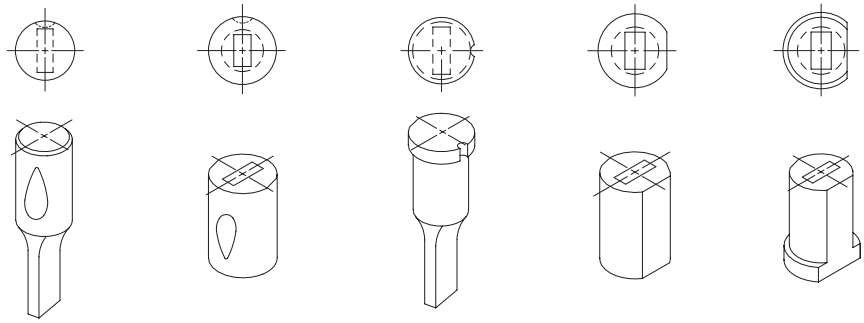
Note:
(1) and (2) - roundings and sharp edges
(1) rounding on the cutting punch of max. R0.2, corresponds to a sharp edge on the matrix (2) rounding on the cutting matrix of max. R0.2, corresponds to a sharp edge on the punch



KEYING INFORMATION - OPTIONS (1/2)

STANDARD LOCATION

BALL LOCK PUNCH | BALL LOCK MATRIX | HEADED PUNCH | MATRIX STRAIGHT | MATRIX HEADED



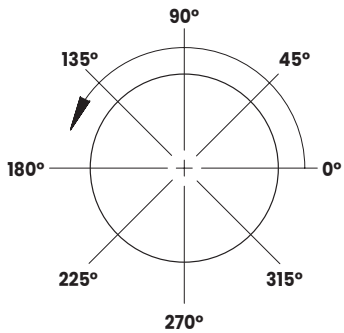
Standard ballseat location for all ball lock products is 90°. Standard flat and dowel groove location is at 0°.

Note: 0° is at 3:00. Alternate no charge locations are 0°, 90°, 180°, 270° (see drawing).

To order alternate locations specify **Keying** type / _ _ _°.

Ball lock punch example: BAO x 13 x P8.2 x W6.1 x 19 x 100 - BS/0° **Matrix Straight example:** MKO25-32 P=13.0 W=10.5 - K1

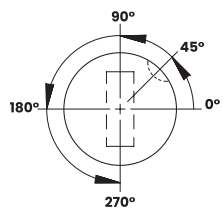
CUSTOM LOCATIONS



TYPICAL EXAMPLES:

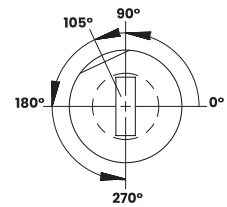
CUSTOM BALL SEAT LOCATION PUNCH

BS/45⁰



CUSTOM FLAT LOCATION MATRIX

K2/105⁰



Any Keying Options can be radially positioned by specifying the appropriate keying and it's desired angle.
The appropriate angle is defined by a counter clockwise rotation from 0°.

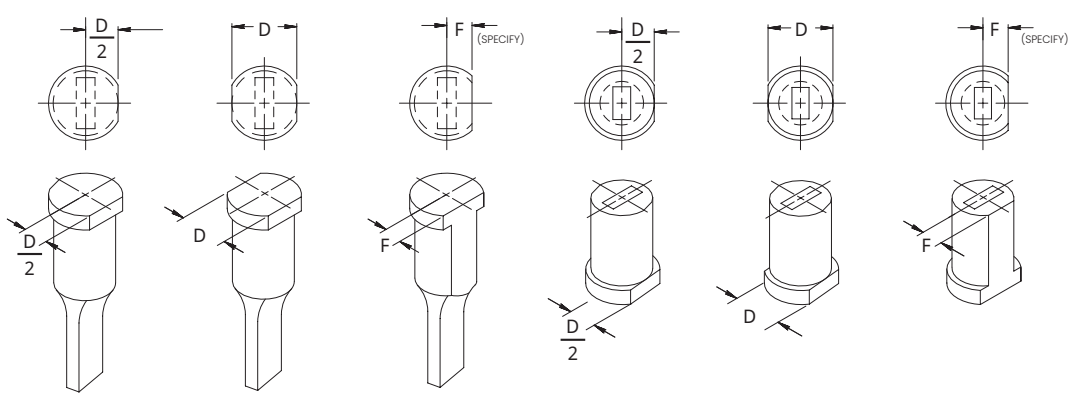
Note: Parts are viewed in die position looking from above the die. Punches are viewed looking through the shanks. Matrixes are viewed through top face.

FLATS ON HEADED PRODUCTS

K1 and K5 are ground flush to shank. **K6** is a user defined flat and requires an **"F"** dimension specified with order.

K5 flats are standard 180° apart.

K1 KEY FLAT | **K5** DOUBLE FLAT | **K6** BODY FLAT | **K1** KEY FLAT | **K5** DOUBLE FLAT | **K6** BODY FLAT



KEYING INFORMATION - OPTIONS (2/2)

FLATS ON MATRIX - STRAIGHT

K1 uses **"F"** dimension as defined in chart, unless an alternate **"F"** dimension is specified with order.
K2 and **K3** use **"F"** dimensions as defined in chart. **K4** and **KX** are user defined flats and requires **"F"** and **"U"** dimension specified with order.

K1 KEY FLAT

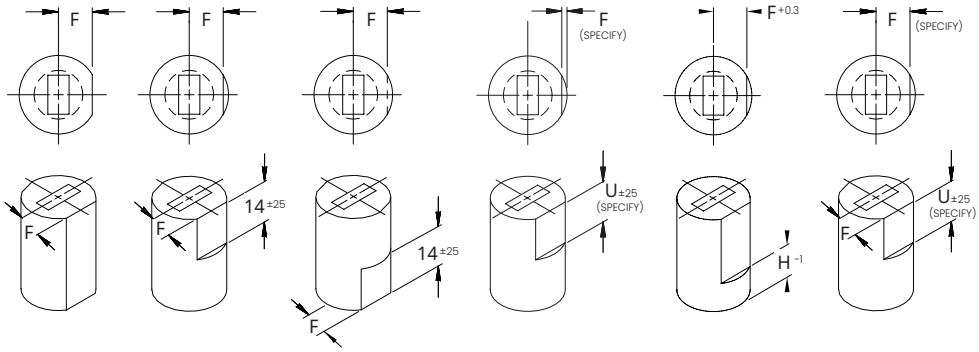
K2 STEP FLAT TOP

K3 STEP FLAT BOTTOM

K4 USER DEFINED FLAT - STYLE 1

K7 STEP FLAT TOP

KX USER DEFINED FLAT - STYLE 2



K1 - K2 - K3		K7	
BODY DIA	"F" +0.02	F	H
8	3.5	-	-
10	4.0	-	-
13	5.5	5.0	9
16	7.0	6.5	9
20	8.5	8.5	9
22	9.5	-	-
25	11.0	10.5	16
32	14.0	14.0	16
38	17.0	-	-
40	18.0	18.0	16
45	20.5	-	-
50	23.0	23.0	16
56	26.0	-	-
63	29.5	29.5	16
71	33.5	-	-
76	35.5	-	-
85	40.0	-	-
90	42.5	-	-
100	47.5	-	-

DOWEL GROOVES

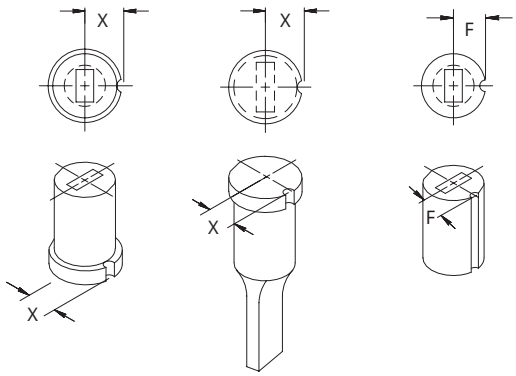
PK3, PK4 and **PK6** are standard **3, 4, 6**, dowel grooves. On Headed products, the groove is positioned tangent to shank. On all other products the dowel is ground to an **"F"** dimension as shown on chart. **PKX** is a user defined dowel groove and requires an **"F"** dimension and dowel size specified with the order.

PK3 - PK4 - PK6

PKX

X= D/2 + DOWEL / 2

X= D/2 + DOWEL / 2



BODY DIA. "D"	PKX SPECIFY DOWEL	PK3 3.0 DOWEL	PK4 4.0 DOWEL	PK6 6.0 DOWEL
	"F" Dimensions			
8	SPECIFY	4.7	5.2	6.2
10	SPECIFY	5.5	6.0	7.0
13	SPECIFY	6.7	7.2	8.2
16	SPECIFY	8.0	8.0	9.0
20	SPECIFY	10.0	10.0	11.0
22	SPECIFY	11.0	11.0	12.0
25	SPECIFY	12.5	12.5	13.5
32 ≥	SPECIFY	D/2	D/2	D/2

SHEAR ANGLES

Angle **"A"** and dimension **"F"** are user defined and must be specified with order.

QS1 BACK TAPER

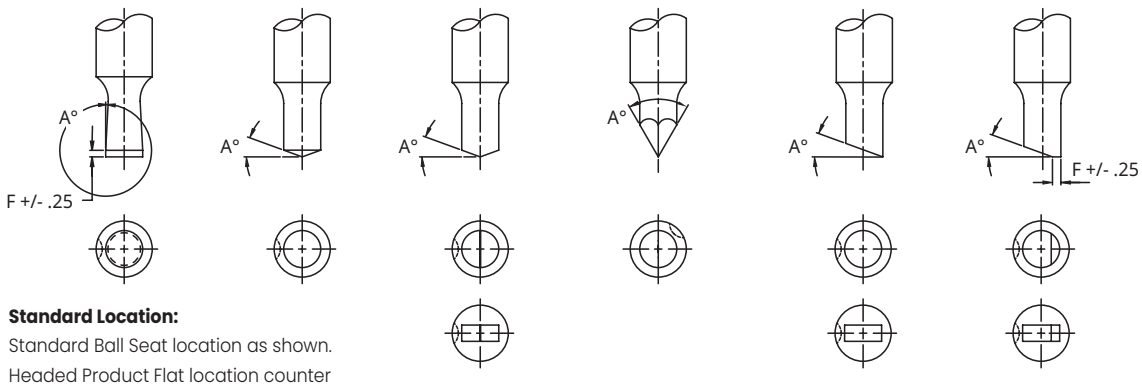
QS2 CONICAL

QS3 ROOF TOP

QS4 NAIL POINT

QS5 SHEAR

QS6 SHEAR WITH FLAT



Standard Location:
Standard Ball Seat location as shown. Headed Product Flat location counter clockwise 90°.

OVERVIEW: PUNCH BLANKS

“ Our ready-to-ship solid blanks from stock “



HEAD SHAPE			M2 (1.3343)		ASP30 (PS / PM)	
			without ejector	with ejector	without ejector	with ejector
Conical Head Punch	h6	DIN 9861 ISO 6752	***HBZ***			
Cylinder Head Punch	m5	ISO 8020	***SBZAPM2***	***SFZAJM2***	***SBZ-APPS***	***SFZ-AJPS***
Heavy Duty Headed Punch	m5		***SBZF***	***SFZF***		
Center Dowel Headed Punch	m5	ISO 8020 with center dowel	***DSZ***	***DFZ***		
Bottle-neck Punch	h6	DIN 5118	***UBZ***	***UFZ***		
Ball Lock Punch Light Duty	g5	ISO 10071	***BHZ***	***BFZ***		
Ball Lock Punch Heavy Duty	g5	NAAMS	***BAZ***	***BAFZ***		

EDM Matrix Blanks - from stock

HEAD SHAPE	Matrix with Counter-bore Relief	Headed, Matrix with Counter-bore Relief
	n5 j6	m5
	ISO 8977	ISO 8977
M2 (1.3343)	***MKB***	***MLB***

Punch Retainers - from stock

Details			Code
Punch Retainer for Round Punches		ISO 8020	***GPR***
Punch Retainer for Shaped Punches		ISO 8020	***GPRS***
Ball Lock Punch Retainer Light Duty		ISO 10071 – NAAMS	***GCRT***
Ball Lock Punch Retainer Heavy Duty		ISO 10071 – NAAMS	***GBRT***
Heavy Duty Headed Punch Retainer for Round Punches			***GPRF***
Heavy Duty Headed Punch Retainer for Shaped Punches			***GPRFS***
Mounting Ring for Bottle-neck Punch		DIN 5118	***G162***



Screw Form Punch & Die Button Unit

Code: **VYZ**

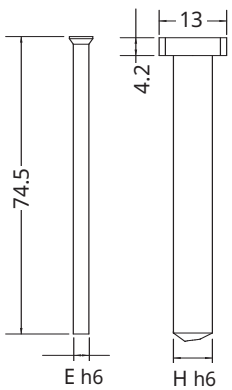
Consists of 3 basic parts.

1- Hss pierce punch
2- Embossing die punch
3- Die button.

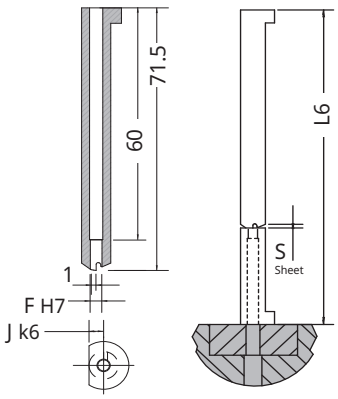




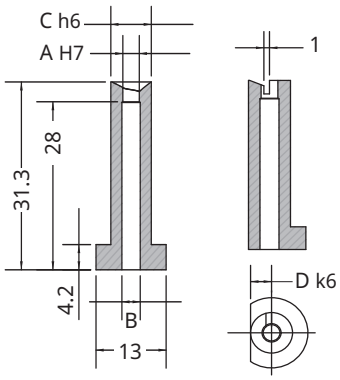
1 - HSS Pierce Punch



2 - Embossing Die Punch



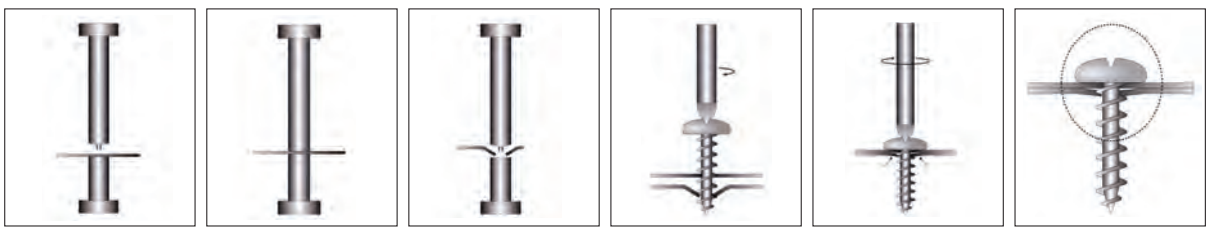
3 -Die Button

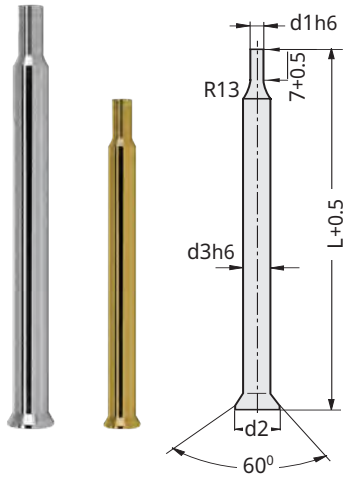


Order Code	Screw	A	B	C	D	E	F	H	J	L6	S Sheet
VYZ B35	B 3.5	2.75	3.2	7.5	3.75	2.7	2.7	7.5	3.75	101.72	0.5
VYZ B39	B 3.9	3.05	3.4	7.5	3.75	3	3	7.5	3.75	101.85	0.63
VYZ B42	B 4.2	3.15	3.5	8.5	4.25	3.1	3.1	8	4	101.97	0.75
VYZ B48	B 4.8	3.85	4.2	9	4.5	3.8	3.8	8	4	102.10	0.88
VYZ B55	B 5.5	4.35	4.8	9	4.5	4.3	4.3	8	4	102.22	1
VYZ B63	B 6.3	4.85	5.3	10.5	5.25	4.8	4.8	10	5	102.42	1.2

In terms of working systematics; in all cutting and forming dies, it creates screw forms with the purpose of connecting two sheet metal strips.
Usage: Hss pierce punch is brought to press operation by inserting in die button. While the embossing on the end of the die button bends and embosses, the HSS pierce punch forms hole. The sheet metals are prepared to be screwed.
The unit forms a hole in screw thread dimension tolerance and helical thread grooves with embossing for screwing and tightening (putting together).
The unspecified tolerances will be according to DIN ISO 2768 T1 (f).

Sheet Metal Thickness: (s)
Screw 3.5mm: Max.0.50 mm. Thickness
Screw 3.9mm: Max.0.63 mm. Thickness
Screw 4.2mm: Max.0.75 mm. Thickness
Screw 4.8mm: Max.0.88 mm. Thickness
Screw 5.5mm: Max.1.00 mm. Thickness
Screw 6.3mm: Max.1.20 mm. Thickness





DIN 9861 **Form C**

Conical Head Punches Stepped

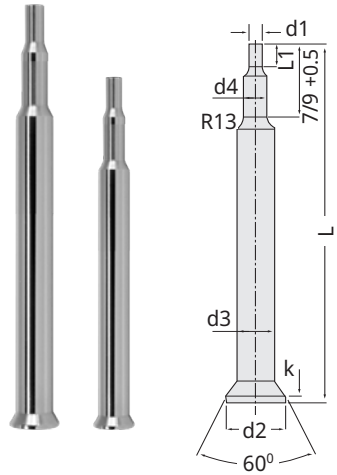
Code: **HKZ**

d1	L	d3	d2
0.8	71	2	3
	80		
1	71		
	80		
1.2	71		
	80		
1.5	71	2.5	
	80		
1.6	71		
	80		
1.8	71		
	80		
1	71	3	4.5
	80		
1.5	71		
	80		
1.8	71		
	80		
2.2	71		
	80		
2.3	71		
	80		
2.6	71		
	80		
2.8	71		
	80		

According to DIN 9861 norm, there is limitation for body and cutting diameter. In this norm, while body diameter for conical head stepped punch is max. 3 mm cutting diameter can be max. 2.95 mm.

Order: **HKZ.** d3 x d1 x L

Material: 1.3343 (M2)
Hardness: 62 - 64 HRC



DIN 9861 **Form E**

Conical Head Punches Stepped

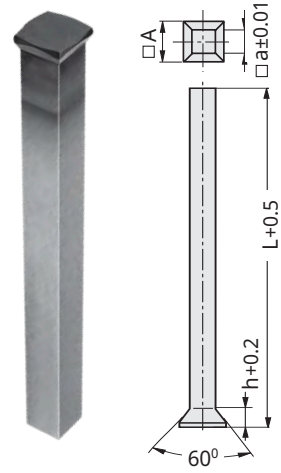
Code: **H2K**

d1	d4	d3	d2	L	L1	k
1	1.4	3	4.5	71 80 100	2	0.5
1.2	1.6					
1.5	1.8					
1.7	2.1					
2.2	2.6					
2.5	2.8					
1.6	2.2	4	5.5	71 80 100	2.5	0.5
1.8	2.3					
2.3	2.8					
2.6	3					
3	3.5					
3.2	3.7					
2.4	3	5	6.5	71 80 100	3	0.5
2.8	3.5					
3.2	4					
3.6	4.2					
4.2	4.6					
4.5	4.8					
2.5	3.2	6	8	71 80 100	3	0.5
3	3.8					
3.5	4.2					
4	4.8					
4.5	5.3					
5	5.5					
5.5	5.8					

It is preferred for thin work pieces and light duty works. Upon request, our punch production is available in the desired material and dimensions and also in shapes.

Order: **H2K.** d1 x d4 x d3 x L x L1

Material: 1.3343 (M2)
Hardness: 62 - 64 HRC



Slot - **Form DA** forged head

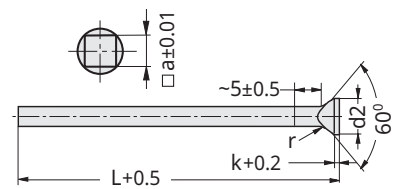
Square Head Punch, Slot Stepped

Code: **HFK**

a	L	A	h
1	71	1.8	1.2
2		3	1.4
3		4.5	1.8
4		5.5	1.8
5		6.5	1.8
6		8	2.2

a	L	A	h
7	71	9	2.8
8		10	
9		11	
10		12	
12		14	

Order: **HFK.** a x L



Conical Head Punch Shaped

Code: **HFY**

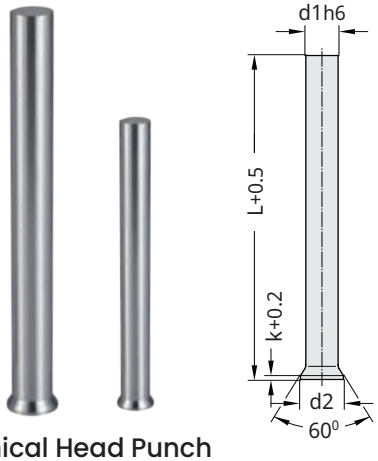
a	L	A	k
1	71	1.8	0.5
2		3	
3		4.5	
4		6	
5		7.5	
6		9	

a	L	A	k
7	71	10.5	1
8		12	
9		13.5	
10		15	
12		18	

Excluding product types in tables, our production such as headless and different slot types, also punches upon request are available.

Order: **HFY.** a x L

HFK **HFY**
Material: 1.3343 (M2)
Hardness: 62 - 64 HRC



Conical Head Punch

DIN 9861 / ISO 6752

Code: **HBZ**

d1	L	d2	k
0.5	71	0.9	0.2
	80		
	100		
0.6	71	1.1	0.2
	80		
	100		
0.7	71	1.3	0.2
	80		
	100		
0.8	71	1.4	0.4
	80		
	100		
0.9	71	1.6	0.4
	80		
	100		
1	71	1.8	0.5
	80		
1.1	100		
1.2	71	2	0.5
	80		
1.3	100		
1.4	71	2.2	0.5
	80		
1.5	100		
1.6	71	2.5	0.5
	80		
1.7	100		
1.8	71	2.8	0.5
	80		
1.9	100		
2	71	3	0.5
	80		
	100		
	125		
	160		
2.1	71	3.2	0.5
	80		
2.2	100		
2.3	71	3.5	0.5
	80		
2.4	100		

d1	L	d2	k
2.5	71	3.5	0.5
	80		
	100		
	125		
	160		
2.6	71	4	0.5
	80		
2.7	100		
2.8	71	4	0.5
	80		
2.9	100		
3	71	4.5	0.5
	80		
	100		
	125		
	160		
3.1	71	4.5	0.5
	80		
3.2	100		
3.3	71	4.5	0.5
	80		
3.4	100		
3.5	71	5	0.5
	80		
	100		
	125		
	160		
3.6	71	5	0.5
	80		
3.7	100		
3.8	71	5	0.5
	80		
3.9	100		
4	71	5.5	0.5
	80		
	100		
	125		
	160		

d1	L	d2	k
4.1	71	5.5	0.5
	80		
4.2	100		
4.3	71	5.5	0.5
	80		
4.4	100		
4.5	71	6	0.5
	80		
	100		
	125		
	160		
4.6	71	6	0.5
	80		
4.7	100		
4.8	71	6	0.5
	80		
4.9	100		
5	71	6.5	0.5
	80		
	100		
	125		
	160		
5.1	71	6.5	0.5
	80		
5.2	100		
5.3	71	6.5	0.5
	80		
5.4	100		
5.5	71	7	0.5
	80		
	100		
	125		
	160		
5.6	71	7	0.5
	80		
5.7	100		
5.8	71	7	0.5
	80		
5.9	100		
6	71	8	0.5
	80		
	100		
	125		
	160		

d1	L	d2	k
6.1	71	8	0.5
	80		
6.2	100		
6.3	71	8	0.5
	80		
6.4	100		
6.5	71	9	1
	80		
	100		
	125		
	160		
6.6	71	9	1
	80		
6.7	100		
6.8	71	9	1
	80		
6.9	100		
7	71	9	1
	80		
	100		
	125		
	160		
7.1	71	9	1
	80		
7.2	100		
7.3	71	9	1
	80		
7.4	100		
7.5	71	10	1
	80		
	100		
	125		
	160		
7.6	71	10	1
	80		
7.7	100		
7.8	71	10	1
	80		
7.9	100		
8	71	10	1
	80		
	100		
	125		
	160		

d1	L	d2	k
8.1	71	10	1
	80		
8.2	100		
8.3	71	10	1
	80		
8.4	100		
8.5	71	11	1
	80		
	100		
	125		
	160		
8.6	71	11	1
	80		
8.7	100		
8.8	71	11	1
	80		
8.9	100		
9	71	11	1
	80		
	100		
9.9	125	12	
	160		
10	71	12	1
	80		
	100		
	125		
	160		
10.5	71	13	1
	80		
	100		
	125		
	160		
11	71	13	1
	80		
	100		
	125		
	160		
11.5	71	14	1
	80		
	100		

d1	L	d2	k
12	71	14	1
	80		
	100		
	125		
	160		
12.5	71	15	1
	80		
	100		
13	71	15	1
	80		
	100		
	125		
	160		
13.5	71	16	1.5
	80		
	100		
14	71	16	1.5
	80		
	100		
	125		
	160		
14.5	71	17	1.5
	80		
	100		
15	80	17	1.5
	100		
	125		
	160		
16	71	18	1.5
	80		
	100		
	125		
	160		
17	80	19	1.5
	100		
18	80	20	1.5
	100		
19	80	21	1.5
	100		
20	71	22	1.5
	80		
	100		
	125		
	160		

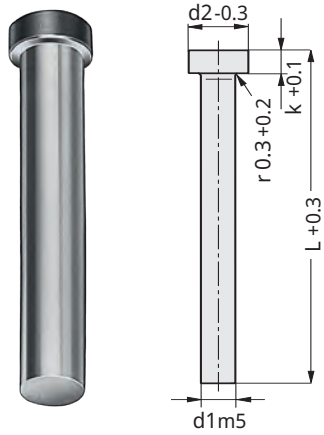
Conical Head Punch - Form D

With our wide range and dimensional products of "Punch and Matrix Sets" from our stocks, we are in top location with our sectoral experience in quality / price formation. In addition, we provide other special requests with the shortest delivery time. Also, full or partial coating can be preferred upon request, by providing resistance against heat and friction on external layer, it solves problems such as winding and cold welding.
For harder work pieces, ASP / PS (powder metal) and for very hard and abrasive work pieces, sintered carbide punches are preferred.

Order:
HBZ. d1 x L

Heat Treated: (60 HRC ± 2)
Surface: ≥ 950 HV 0.3
Head Hardness: (52 ±3 HRC)

Material: < Ø12.5= 1.3343 (M2) Completely ground
≥ Ø12.5= 1.2379 (HWS) Completely ground



Code: **SBZAPM2**

Cylinder Head Punch

Solid Blank - ISO 8020

Order Code	d1	L	d2	k	r
SBZAPM202071	2	71	5	3	0.2
SBZAPM202080		80			
SBZAPM202100		100			
SBZAPM202571	2.5	71	5.5	3	0.2
SBZAPM202580		80			
SBZAPM2025100		100			
SBZAPM203071	3	71	6.5	3	0.2
SBZAPM203080		80			
SBZAPM203100		100			
SBZAPM203571	3.5	71	6	3	0.2
SBZAPM203580		80			
SBZAPM2035100		100			
SBZAPM204071	4	71	7	5	0.3
SBZAPM204080		80			
SBZAPM204100		100			
SBZAPM204571	4.5	71	7.5	5	0.3
SBZAPM204580		80			
SBZAPM2045100		100			
SBZAPM205071	5	71	8	5	0.3
SBZAPM205080		80			
SBZAPM205090		90			
SBZAPM205100		100			
SBZAPM205125		125			
SBZAPM2055100	5.5	100	8.5	5	0.3

Order Code	d1	L	d2	k	r
SBZAPM206071	6	71	9	5	0.3
SBZAPM206080		80			
SBZAPM206100		100			
SBZAPM206125		125			
SBZAPM206160		160			
SBZAPM2065100	6.5	100	9.5	5	0.3
SBZAPM207080	7	80	10	5	0.3
SBZAPM207100		100			
SBZAPM207125		125			
SBZAPM2075100	7.5	100	10.5	5	0.3
SBZAPM208071	8	71	11	5	0.3
SBZAPM208080		80			
SBZAPM208100		100			
SBZAPM208125		125			
SBZAPM208160		160			
SBZAPM208200		200			
SBZAPM2085100	8.5	100	11.5	5	0.3
SBZAPM209080	9	80	12	5	0.3
SBZAPM209100		100			
SBZAPM209125		125			
SBZAPM2095100	9.5	100	12.5	5	0.3
SBZAPM210071	10	71	13	5	0.3
SBZAPM210080		80			
SBZAPM210100		100			
SBZAPM210125		125			
SBZAPM210160		160			
SBZAPM210200		200			
SBZAPM2105100	10.5	100	13.5	5	0.3
SBZAPM211080	11	80	14	5	0.3
SBZAPM211100		100			
SBZAPM211125		125			
SBZAPM2115100	11.5	100	14.5	5	0.3
SBZAPM212071	12	71	15	5	0.3
SBZAPM212080		80			
SBZAPM212100		100			
SBZAPM212125		125			
SBZAPM212160		160			
SBZAPM2125100	12.5	100	15.5	5	0.3
SBZAPM213071	13	71	16	5	0.3
SBZAPM213080		80			
SBZAPM213090		90			
SBZAPM213100		100			
SBZAPM213125		125			
SBZAPM213160		160			
SBZAPM213200		200			
SBZAPM2135100	13.5	100	16.5	5	0.3

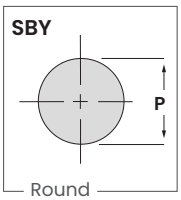
Order Code	d1	L	d2	k	r
SBZAPM214080	14	80	17	5	0.3
SBZAPM214100		100			
SBZAPM214125		125			
SBZAPM2145100	14.5	100	17.5	5	0.3
SBZAPM215100	15	100	18	5	0.3
SBZAPM215125		125			
SBZAPM2155100	15.5	100	18.5	5	0.3
SBZAPM216071	16	71	19	5	0.3
SBZAPM216080		80			
SBZAPM216090		90			
SBZAPM216100		100			
SBZAPM216125		125			
SBZAPM216160		160			
SBZAPM216200		200			
SBZAPM2165100	16.5	100	19.5	5	0.3
SBZAPM217100	17	100	20	5	0.3
SBZAPM2175100	17.5	100	20.5	5	0.3
SBZAPM218100	18	100	21	5	0.3
SBZAPM218125		125			
SBZAPM219100	19	100	22	5	0.3
SBZAPM220071	20	71	23	5	0.3
SBZAPM220080		80			
SBZAPM220090		90			
SBZAPM220100		100			
SBZAPM220125		125			
SBZAPM220160		160			
SBZAPM220200		200			
SBZAPM221100	21	100	24	5	0.3
SBZAPM222100	22	100	25	5	0.3
SBZAPM223100	23	100	26	5	0.3
SBZAPM224100	24	100	27	5	0.3
SBZAPM225071	25	71	28	5	0.3
SBZAPM225080		80			
SBZAPM225090		90			
SBZAPM225100		100			
SBZAPM225125		125			
SBZAPM225160		160			
SBZAPM225200		200			
SBZAPM226100	26	100	29	5	0.3
SBZAPM227100	27	100	30	5	0.3
SBZAPM228100	28	100	31	5	0.3
SBZAPM229100	29	100	32	5	0.3
SBZAPM230100	30	100	33	5	0.3
SBZAPM230125		125			
SBZAPM232080	32	80	35	5	0.3
SBZAPM232090		90			
SBZAPM232100		100			
SBZAPM232125		125			
SBZAPM232160		160			
SBZAPM240100	40	100	43	5	0.3
SBZAPM240125		125			

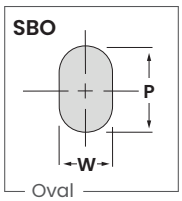
Material: 1.3343 (M2) completely ground
Head Hardness: (52 ±3 HRC)

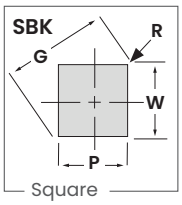
Heat Treated: 62 - 64 HRC **Surface:** ≥ 950 HV 0.3

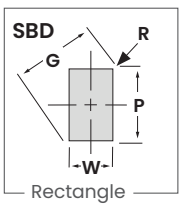
Mounting of cylinder head type punch to the holder plate during usage is planned, guiding process is provided by the stripper plate.
By mounting punches in this style, the elimination of axial errors that resulted from incorrect mounting of die set or press is more easier.
The use of this mounting method provided the alteration between the transmission of punching power and bearing. Also, full or partial coating can be preferred upon request, by providing resistance against heat and friction on external layer, it solves problems such as winding and cold welding.
For harder work pieces, ASP / PS (powder metal) and for very hard and abrasive work pieces, sintered carbide punches are preferred.

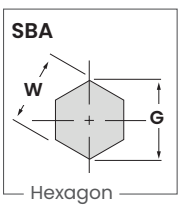
SHAPES

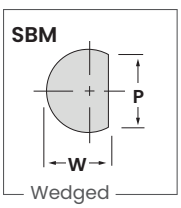


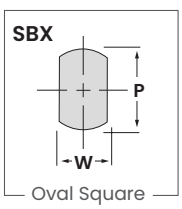


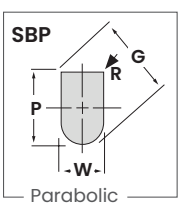




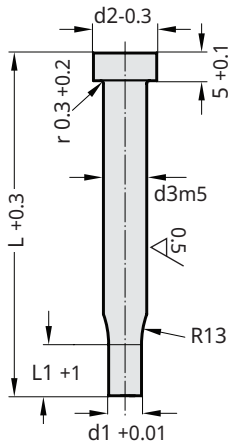




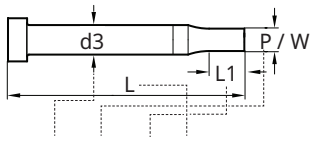








How to order:
Shape
d3
P/W
L1
L
Material



SBY x 10 x P7.0 x 13 x 80 - M2
SBO x 10 x P8.2 x W5.2 x 25 x 100 - M2

Cylinder Head Stepped Punch / ISO 8020

Code: **SB..**

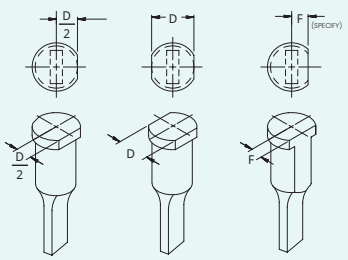
Apart from our standards, special dimensions and complex geometries can be produced according to your technical drawings.

Please refer to the table on the left side

SBY - SBO - SBK - SBD - SBA - SBM - SBX - SBP

Order	d3	Head d2	d1 / Shape (SBY) Round P	d1 / Shape Other Shapes W G/P	Standard L1	Alternative L1 Min. Max.	L
SB.. 04	4	7	1.6 ~ 3.95	1.6 - 4	8	10 - 13	50 63 71 80 90 100 125
SB.. 05	5	8	1.6 ~ 4.95	1.6 - 4.95	13	10 - 19	
SB.. 06	6	9	2.4 ~ 5.95	2.4 - 5.95	13	10 - 19	
SB.. 08	8	11	3.2 ~ 7.95	3.2 - 7.95	19	13 - 25	
SB.. 10	10	13	4.5 ~ 9.95	4.5 - 9.95	19	13 - 25	
SB.. 13	13	16	6 ~ 12.95	6 - 12.95	19	13 - 25	
SB.. 16	16	19	8 ~ 15.95	7.5 - 15.95	19	13 - 25	
SB.. 20	20	23	10 ~ 19.95	8 - 19.95	19	13 - 25	71 80 90 100 125
SB.. 25	25	28	12 ~ 24.95	9 - 24.95	19	13 - 25	
SB.. 32	32	35	16 ~ 31.95	10 - 31.95	25	19 - 30	
SB.. 40	40	43	20 ~ 39.95	14 - 39.95	25	19 - 30	

FLATS ON HEADED PRODUCTS



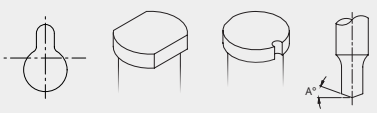
K1 and K5 are ground flush to shank. **K6** is a user defined flat and requires an **"F"** dimension specified with order.
K5 flats are standard 180° apart.

Order Example with Flat:
SBO x 10 x P8.2 x W5.2 x 25 x 100 - **K1** - M2
Keying Info

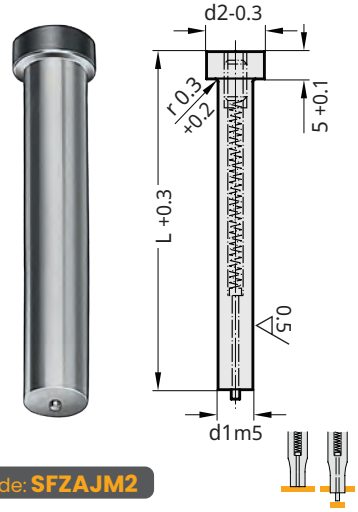
Order Example with Flat and Coating:
SBO x 10 x P8.2 x W5.2 x 25 x 100 - K1 - M2 - **Alcrn**
Coating Info

OPTIONAL FEATURES

COATING



For details on Keying Information / Key Flat Options, More Complex Special Shapes, Shear Angles, and Coatings, please refer to the beginning of Category 6.



Code: **SFZAJM2**

Cylinder Head Punch with Ejector

Solid Blank - ISO 8020

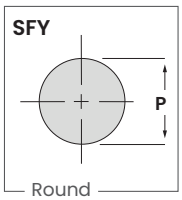
Order Code	d1	L	d2
SFZAJM205050	5	50	8
SFZAJM205063		63	
SFZAJM205071		71	
SFZAJM205080		80	
SFZAJM205090		90	
SFZAJM205100		100	
SFZAJM205125		125	
SFZAJM255100	5.5	100	8.5
SFZAJM206050	6	50	9
SFZAJM206063		63	
SFZAJM206071		71	
SFZAJM206080		80	
SFZAJM206090		90	
SFZAJM206100		100	
SFZAJM206125		125	
SFZAJM2065100	6.5	100	9.5
SFZAJM207100	7	100	10
SFZAJM2075100	7.5	100	10.5
SFZAJM208050	8	50	11
SFZAJM208063		63	
SFZAJM208071		71	
SFZAJM208080		80	
SFZAJM208090		90	
SFZAJM208100		100	
SFZAJM208125		125	
SFZAJM208160		160	
SFZAJM2085100	8.5	100	11.5
SFZAJM209100	9	100	12
SFZAJM2095100	9.5	100	12.5
SFZAJM210050	10	50	13
SFZAJM210063		63	
SFZAJM210071		71	
SFZAJM210080		80	
SFZAJM210090		90	
SFZAJM210100		100	
SFZAJM210125		125	
SFZAJM210160		160	
SFZAJM2105100	10.5	100	13.5
SFZAJM211080	11	80	14
SFZAJM211100		100	
SFZAJM2115100	11.5	100	14.5
SFZAJM212050	12	50	15
SFZAJM212063		63	
SFZAJM212071		71	
SFZAJM212080		80	
SFZAJM212090		90	
SFZAJM212100		100	
SFZAJM212125		125	
SFZAJM212160		160	
SFZAJM2125100	12.5	100	15.5
SFZAJM213050	13	50	16
SFZAJM213063		63	
SFZAJM213071		71	
SFZAJM213080		80	
SFZAJM213090		90	
SFZAJM213100		100	
SFZAJM213125		125	
SFZAJM213160		160	
SFZAJM2135100	13.5	100	16.5
SFZAJM214080	14	80	17
SFZAJM214100		100	
SFZAJM2145100	14.5	100	17.5
SFZAJM215080	15	80	18
SFZAJM215100		100	
SFZAJM2155100	15.5	100	18.5
SFZAJM216050	16	50	19
SFZAJM216063		63	
SFZAJM216071		71	
SFZAJM216080		80	
SFZAJM216090		90	
SFZAJM216100		100	
SFZAJM216125		125	
SFZAJM216160		160	
SFZAJM2165100	16.5	100	19.5
SFZAJM217100	17	100	20
SFZAJM218100	18	100	21
SFZAJM219100	19	100	22
SFZAJM220063	20	63	23
SFZAJM220071		71	
SFZAJM220080		80	
SFZAJM220090		90	
SFZAJM220100		100	
SFZAJM220125		125	
SFZAJM220160		160	
SFZAJM221100	21	100	24
SFZAJM222100	22	100	25
SFZAJM223100	23	100	26
SFZAJM224100	24	100	27
SFZAJM225063	25	63	28
SFZAJM225071		71	
SFZAJM225080		80	
SFZAJM225090		90	
SFZAJM225100		100	
SFZAJM225125		125	
SFZAJM225160		160	
SFZAJM226100	26	100	29
SFZAJM227100	27	100	30
SFZAJM228100	28	100	31
SFZAJM229100	29	100	32
SFZAJM230100	30	100	33
SFZAJM232090	32	90	35
SFZAJM232100		100	
SFZAJM232125		125	
SFZAJM232160		160	
SFZAJM240100	40	100	43
SFZAJM240125		125	

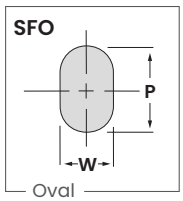
Material: 1.3343 (M2) completely ground
Head Hardness: (52 ±3 HRC)

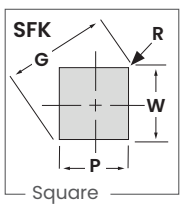
Heat Treated: 62 - 64 HRC **Surface:** ≥ 950 HV 0.3

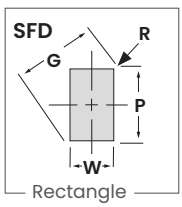
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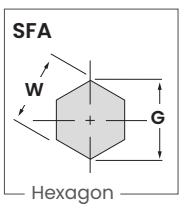
SHAPES

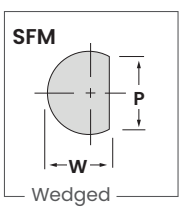


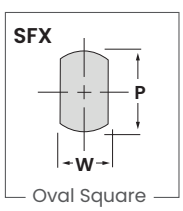


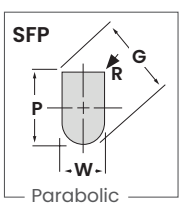




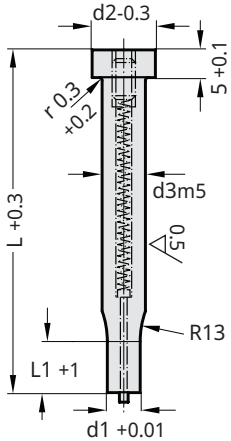




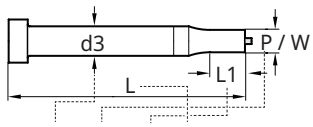








How to order:
Shape
d3
P/W
L1
L
Material

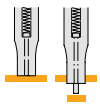


SFY x 10 x P7.0 x 13 x 80 - M2
SFO x 10 x P8.2 x W5.2 x 25 x 100 - M2

Cylinder Head Stepped Punch with Ejector

Code: **SF..**

Apart from our standards, special dimensions and complex geometries can be produced according to your technical drawings.



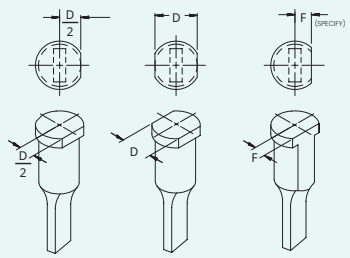
Please refer to the table on the left side

SFY - SFO - SFK - SFD - SFA - SFM - SFX - SFP

Order	d3	Head d2	d1 / Shape (SFY) Round P	d1 / Shape Other Shapes W G/P	Standard L1	Alternative L1 Min. Max.	L
SF.. 05	5	8	1.6 ~ 4.95	1.6 - 4.95	13	10 - 19	50
SF.. 06	6	9	2.4 ~ 5.95	2.4 - 5.95	13	10 - 19	
SF.. 08	8	11	3.2 ~ 7.95	3.2 - 7.95	19	13 - 25	63
SF.. 10	10	13	4.5 ~ 9.95	4.5 - 9.95	19	13 - 25	71
SF.. 13	13	16	6 ~ 12.95	6 - 12.95	19	13 - 25	80
SF.. 16	16	19	8 ~ 15.95	7.5 - 15.95	19	13 - 25	
SF.. 20	20	23	10 ~ 19.95	8 - 19.95	19	13 - 25	90
SF.. 25	25	28	12 ~ 24.95	9 - 24.95	19	13 - 25	100
SF.. 32	32	35	16 ~ 31.95	10 - 31.95	25	19 - 30	125
SF.. 40	40	43	20 ~ 39.95	14 - 39.95	25	19 - 30	

FLATS ON HEADED PRODUCTS

K1 KEY FLAT | **K5** DOUBLE FLAT | **K6** BODY FLAT



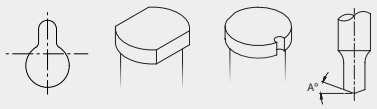
K1 and K5 are ground flush to shank. **K6** is a user defined flat and requires an **"F"** dimension specified with order. **K5** flats are standard 180° apart.

Order Example with Flat:
SFO x 10 x P8.2 x W5.2 x 25 x 100 - **K1** - M2
Keying Info

Order Example with Flat and Coating:
SFO x 10 x P8.2 x W5.2 x 25 x 100 - K1 - M2 - **AlCrN**
Coating Info

OPTIONAL FEATURES

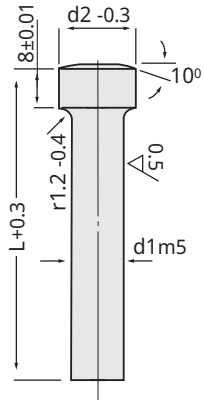
COATING



For details on Keying Information / Key Flat Options, More Complex Special Shapes, Shear Angles, and Coatings, please refer to the beginning of Category 6.

NEW





Heavy Duty Headed Punch

* Larger-headed

Code: **SBZF**

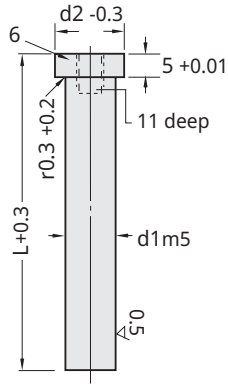
Blank Type

d1	Order			d2
	L: 70	L: 80	L: 100	
6	SBZF 6 x 70	SBZF 6 x 80	SBZF 6 x 100	11
8	SBZF 8 x 70	SBZF 8 x 80	SBZF 8 x 100	13
10	SBZF 10 x 70	SBZF 10 x 80	SBZF 10 x 100	15
12	SBZF 12 x 70	SBZF 12 x 80	SBZF 12 x 100	17
13	SBZF 13 x 70	SBZF 13 x 80	SBZF 13 x 100	18
16	SBZF 16 x 70	SBZF 16 x 80	SBZF 16 x 100	21
20	SBZF 20 x 70	SBZF 20 x 80	SBZF 20 x 100	25
25	SBZF 25 x 70	SBZF 25 x 80	SBZF 25 x 100	30
32	-	-	SBZF 32 x 100	37

NEW







Center Dowel Headed Punch

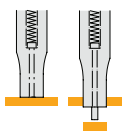
ISO 8020, with center dowel

Code: **DSZ**

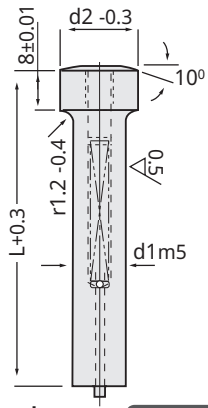
Blank Type

d1	Order				d2
	L: 71	L: 80	L: 90	L: 100	
10	DSZ10 x 71	DSZ10 x 80	DSZ10 x 90	DSZ10 x 100	13
13	DSZ13 x 71	DSZ13 x 80	DSZ13 x 90	DSZ13 x 100	16
16	DSZ16 x 71	DSZ16 x 80	DSZ16 x 90	DSZ16 x 100	19
20	DSZ20 x 71	DSZ20 x 80	DSZ20 x 90	DSZ20 x 100	23
25	DSZ25 x 71	DSZ25 x 80	DSZ25 x 90	DSZ25 x 100	28
32	DSZ32 x 71	DSZ32 x 80	DSZ32 x 90	DSZ32 x 100	35

NEW







Heavy Duty Headed Punch with Ejector

* Larger-headed

Code: **SFZF**

Blank Type

d1	Order			d2
	L: 70	L: 80	L: 100	
8	SFZF 8 x 70	SFZF 8 x 80	SFZF 8 x 100	13
10	SFZF 10 x 70	SFZF 10 x 80	SFZF 10 x 100	15
12	-	-	SFZF 12 x 100	17
13	SFZF 13 x 70	SFZF 13 x 80	SFZF 13 x 100	18
16	SFZF 16 x 70	SFZF 16 x 80	SFZF 16 x 100	21
20	SFZF 20 x 70	SFZF 20 x 80	SFZF 20 x 100	25
25	SFZF 25 x 70	SFZF 25 x 80	SFZF 25 x 100	30
32	-	-	SFZF 32 x 100	37

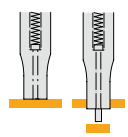
SBZF **SFZF**

Material: 1.3343 (M2) completely ground

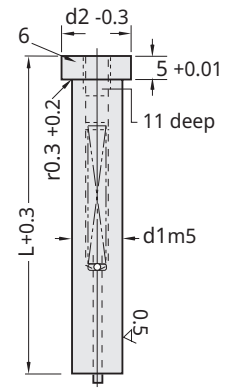
Heat Treated: 60 - 63 HRC / **Head Hardness:** 40 - 55 HRC

NEW









Center Dowel Headed Punch with Ejector

ISO 8020, with center dowel

Code: **DFZ**

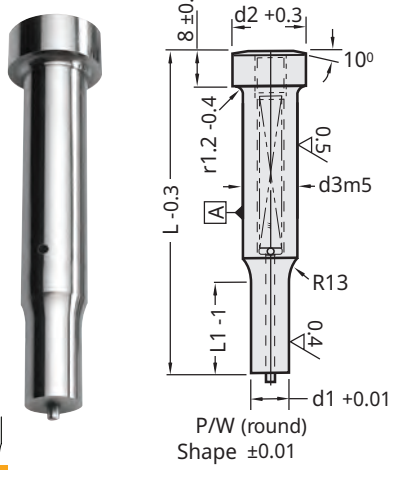
Blank Type

d1	Order				d2
	L: 71	L: 80	L: 90	L: 100	
10	DFZ10 x 71	DFZ10 x 80	DFZ10 x 90	DFZ10 x 100	13
13	DFZ13 x 71	DFZ13 x 80	DFZ13 x 90	DFZ13 x 100	16
16	DFZ16 x 71	DFZ16 x 80	DFZ16 x 90	DFZ16 x 100	19
20	DFZ20 x 71	DFZ20 x 80	DFZ20 x 90	DFZ20 x 100	23
25	DFZ25 x 71	DFZ25 x 80	DFZ25 x 90	DFZ25 x 100	28
32	DFZ32 x 71	DFZ32 x 80	DFZ32 x 90	DFZ32 x 100	35

DSZ **DFZ**

Material: 1.3343 (M2) completely ground

Heat Treated: 60 - 63 HRC / **Head Hardness:** 40 - 55 HRC



Heavy Duty Stepped Punch with Ejector

* Larger-headed

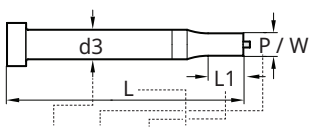
Code: **SF..F**

Please refer to the table on the right side

SFYF-SFOF-SFKF-SFDF-SFAF-SFMF-SFCF-SFPF

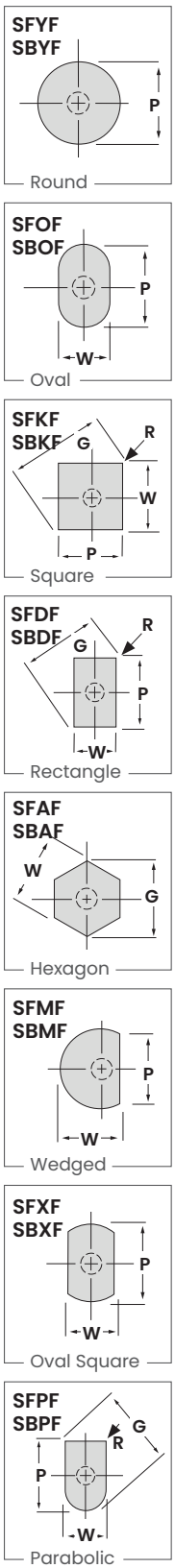
Order d3	Head d2	d1 / Shape (SFYF) Round P	d1 / Shape Other Shapes W G/P	Standard L1	Alternative L1 Min. Max.	L
SF..F08	13	3.2 ~ 7.95	3.2 - 7.95	13	13 25	
SF..F10	15	4.5 ~ 9.95	4.5 - 9.95	13	13 25	
SF..F13	18	6 ~ 12.95	6 - 12.95	13	13 25	70
SF..F16	21	8 ~ 15.95	7.5 - 15.95	19	13 25	80 90
SF..F20	25	10 ~ 19.95	8 - 19.95	19	13 25	100
SF..F25	30	12 ~ 24.95	9 - 24.95	19	13 25	
SF..F32	37	16 ~ 31.95	10 - 31.95	25	19 30	

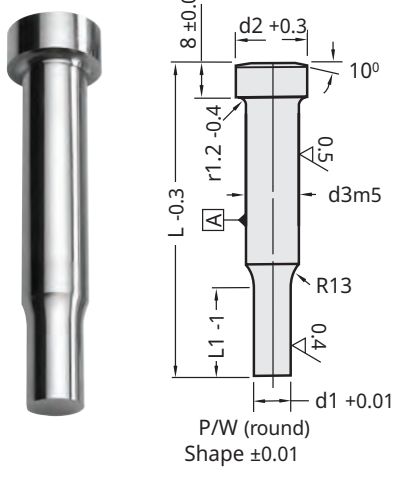
How to order:
Shape
d3
P/W
L1
L
Material



SFYF x 10 x P7.0 x 13 x 80 - M2
SFOF x 10 x P8.2 x W5.2 x 25 x 100 - M2

SHAPES





Heavy Duty Stepped Punch

* Larger-headed

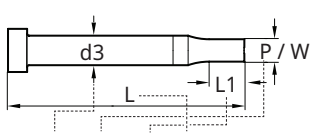
Code: **SB..F**

Please refer to the table on the left side

SBYF-SBOF-SBKF-SBDF-SBAF-SBMF-SBCF-SBPF

Order d3	Head d2	d1 / Shape (SBYF) Round P	d1 / Shape Other Shapes W G/P	Standard L1	Alternative L1 Min. Max.	L
SB..F08	13	3.2 ~ 7.95	3.2 - 7.95	13	13 25	
SB..F10	15	4.5 ~ 9.95	4.5 - 9.95	13	13 25	
SB..F13	18	6 ~ 12.95	6 - 12.95	13	13 25	60 70
SB..F16	21	8 ~ 15.95	7.5 - 15.95	19	13 25	80 90
SB..F20	25	10 ~ 19.95	8 - 19.95	19	13 25	100
SB..F25	30	12 ~ 24.95	9 - 24.95	19	13 25	
SB..F32	37	16 ~ 31.95	10 - 31.95	25	19 30	

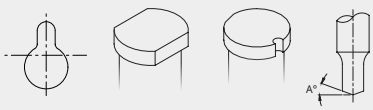
How to order:
Shape
d3
P/W
L1
L
Material



SBYF x 10 x P7.0 x 13 x 80 - M2
SBOF x 10 x P8.2 x W5.2 x 25 x 100 - M2

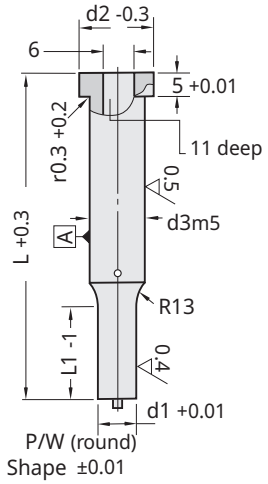
OPTIONAL FEATURES

COATING



For details on Keying Information / Key Flat Options, More Complex Special Shapes, Shear Angles, and Coatings, please refer to the beginning of Category 6.





Center Dowel Stepped Punch with Ejector

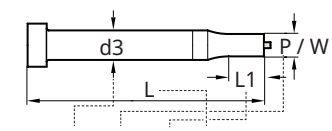
Code: **DF...**

Please refer to the table on the right side

DFY - DFO - DFK - DFD - DFA - DFM - DFX - DFP

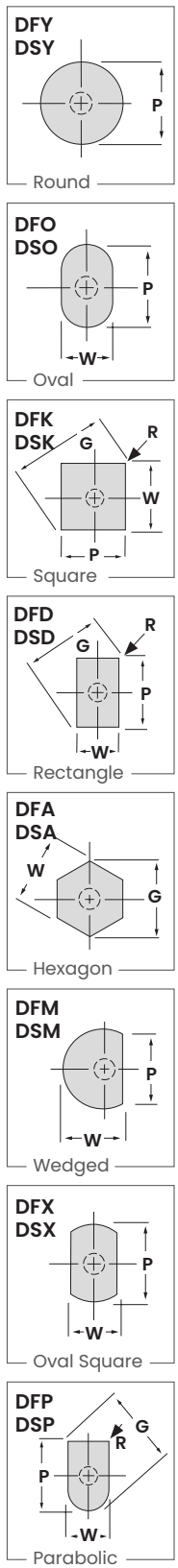
Order d3	Head d2	d1 / Shape (DFY) Round P	d1 / Shape Other Shapes W G/P	Standard L1	Alternative L1 Min. Max.	L
DF.. 10	13	4.5 ~ 9.95	4.5 - 9.95	19	13 25	63
DF.. 13	16	6 ~ 12.95	6 - 12.95	19	13 25	71
DF.. 16	19	8 ~ 15.95	7.5 - 15.95	19	13 25	80
DF.. 20	23	10 ~ 19.95	8 - 19.95	19	13 25	90
DF.. 25	28	12 ~ 24.95	9 - 24.95	19	13 25	100
DF.. 32	35	16 ~ 31.95	10 - 31.95	25	19 30	

How to order:
Shape
d3
P/W
L1
L
Material

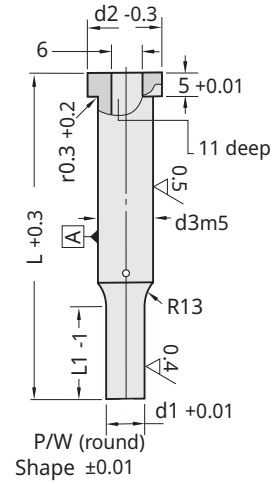


DFY x 10 x P7.0 x 13 x 80 - M2
DFO x 10 x P8.2 x W5.2 x 25 x 100 - M2

SHAPES







Center Dowel Stepped Punch

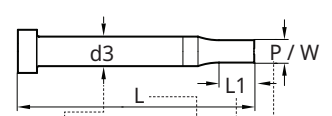
Code: **DS...**

Please refer to the table on the left side

DSY - DSO - DSK - DSD - DSA - DSM - DSX - DSP

Order d3	Head d2	d1 / Shape (DSY) Round P	d1 / Shape Other Shapes W G/P	Standard L1	Alternative L1 Min. Max.	L
DS.. 10	13	4.5 ~ 9.95	4.5 - 9.95	19	13 25	63
DS.. 13	16	6 ~ 12.95	6 - 12.95	19	13 25	71
DS.. 16	19	8 ~ 15.95	7.5 - 15.95	19	13 25	80
DS.. 20	23	10 ~ 19.95	8 - 19.95	19	13 25	90
DS.. 25	28	12 ~ 24.95	9 - 24.95	19	13 25	100
DS.. 32	35	16 ~ 31.95	10 - 31.95	25	19 30	

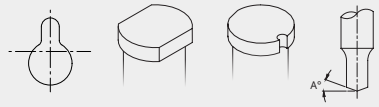
How to order:
Shape
d3
P/W
L1
L
Material



DSY x 10 x P7.0 x 13 x 80 - M2
DSO x 10 x P8.2 x W5.2 x 25 x 100 - M2

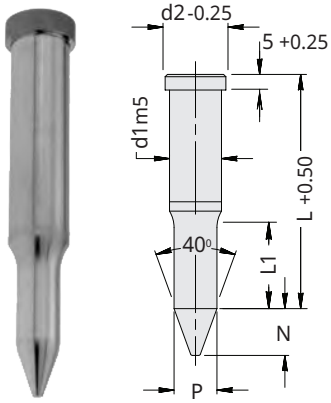
OPTIONAL FEATURES

COATING



For details on Keying Information / Key Flat Options,
More Complex Special Shapes, Shear Angles, and Coatings,
please refer to the beginning of Category 6.

Pilots enter a previously punched hole to accurately position the workpiece and hold it in place before the main punches operate. This ensures precise and repeatable punching operations.

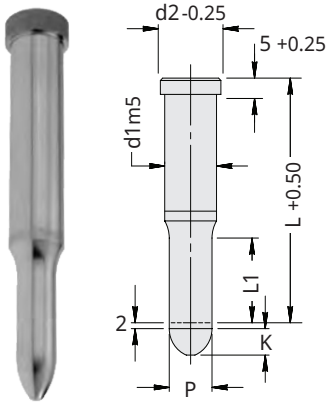


Code: **SPK**

Pilot with Taper End

d1	d2	Taper P	N	Ref P	L1 Standard	L1 Alternative	L
10	13	4.85 ~ 10	8	5.6	19	25	71
13	16	6.30 ~ 13	10	7.1			80
16	19	9.95 ~ 16	15	10.8			100
20	23	13.6 ~ 20	20	14.4			125
25	28	17.3 ~ 25	25	18			140
32	35	20.9 ~ 32	30	21.7	25	30	

Order: **SPK** d1 x P x L1 x L

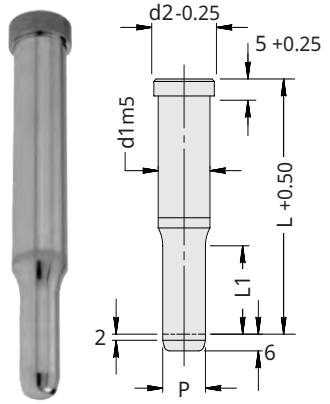


Code: **SPX**

Pilot - Spherical

d1	d2	K	Taper P	Standar d L1	Alternative L1 min.	Alternative L1 max.	L
4	7	4	1.55 ~ 4	8	10	-	50
5	8		1.55 ~ 5	13			60
6	9		1.55 ~ 6	13			71
8	11		2.45 ~ 8	19	13	13	80
10	13	10	3.15 ~ 10				100
13	16		4.95 ~ 13				
16	19		7.95 ~ 16				
20	23		9.95 ~ 20				
25	28		11.95 ~ 25				
32	35		15.95 ~ 32	25	19	30	

Order: **SPX** d1 x P x L1 x L



Code: **SPP**

Pilot - Parabolic Type

Order: **SPP** d1 x P x L1 x L

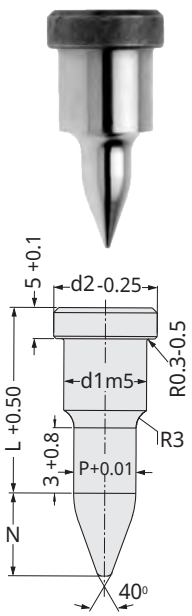
Standard Tolerances

Round / **P** +01 -00 | .01 | From **P** to **d1**

When P = d1, shank / body tolerances apply.

SPK - SPX - SPP

Material: 1.3343 (M2)
Hardness: 62 - 64 HRC



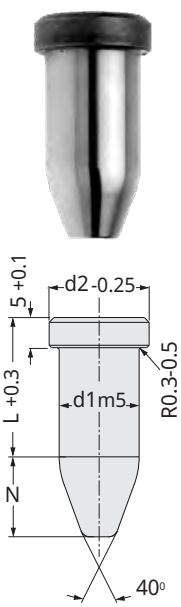
Code: **SMK** **Pointed End**

Pilot - Compact Type

d1	P	N	d2	L
4	1.95 ~ 3.99	4	7	20
5	2.65 ~ 4.99	5	8	22
6	3.3 ~ 5.99	6	9	25
8	4.1 ~ 7.99	7	11	28
10	4.8 ~ 9.99	8	13	35
13	6.25 ~ 12.99	10	16	
16	9.85 ~ 15.99	15	19	
20	13.5 ~ 19.99	20	23	
25	17.2 ~ 24.99	25	28	
32	20.8 ~ 31.99	30	35	

Order: **SMK** d1 x P x N x L

Material: 1.3343 (M2)
Hardness: 62 - 64 HRC



Code: **SMX** **Spherical End**

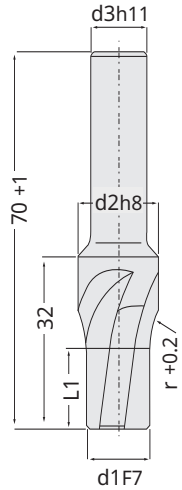
Pilot - Compact & Spherical

d1	N	d2	L
3.01 ~ 4	4	7	20
4.01 ~ 5	5	8	22
5.01 ~ 6	6	9	25
6.01 ~ 8	7	11	28
8.01 ~ 10	8	13	35
10.01 ~ 13	10	16	
13.01 ~ 16	15	19	
16.01 ~ 20	20	23	

Order: **SMX** d1 x N x L

Material: 1.3343 (M2)
Hardness: 62 - 64 HRC





Code: **UFT**

Piloted Counterbore Punch

(for tapered head)

d1	d2	d3	r	L1	x (d1)
2	3.3	3.5	3.5	5	01
2.1 ~ 2.2	3.5	3.5	5	5	01
2.3 ~ 2.5	3.8	4.0	5	5	01
2.6 ~ 2.9	4.3	4.5	6	7	01
3 ~ 3.4	4.9	5.0	6	7	01
3.5 ~ 3.9	5.4	5.5	8	7	01
4 ~ 4.4	5.9	6.0	8	8	01
4.5 ~ 4.9	6.4	6.5	8	8	01
5 ~ 5.4	7.4	7.5	10	10	01
5.5 ~ 5.9	8.5	8.5	10	10	01
6	9.5	10	12	12	05
6.5 ~ 7	10.5	10	12	12	05
7.5 ~ 8	11.5	10	12	12	05
8.5 ~ 9	13.5	10	15	12	05
9.5 ~ 10	14.5	10	15	12	05
10.5 ~ 11	15.5	10	15	15	05
11.5 ~ 12	16.5	10	15	15	05
12.5 ~ 13	17.5	10	15	15	05
13.5 ~ 14	18.5	10	15	15	05
14.5 ~ 15	19.5	10	15	15	05
15.5 ~ 16	20.5	10	15	15	05
16.5 ~ 17	21.5	16	15	15	05
17.5 ~ 18	22.5	16	15	15	05
18.5 ~ 19	23.5	16	15	15	05
19.5 ~ 20	25.5	16	15	15	05

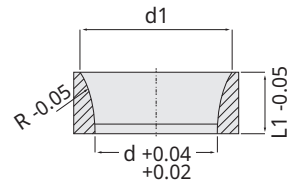
Order: **UFT.** d1

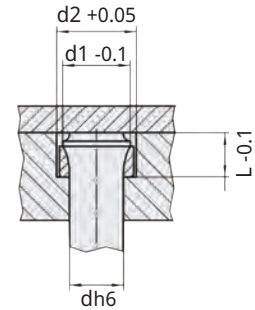
Material: 1.3343 (M2)
Hardness: 62 - 66 HRC



Code: **G162**

Mounting Ring for Bottle-neck Punch

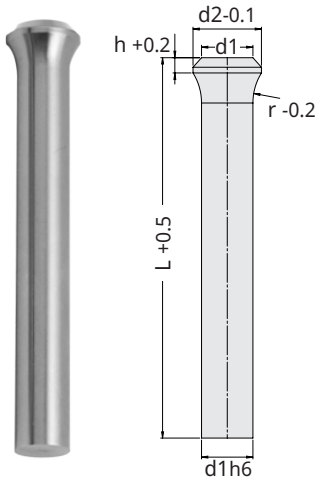




Order Code	d	d1	d2	L	L1	R
G162 30	3	4.5	6	7	4	6.5
G162 35	3.5	5	7	7	4	8
G162 40	4	5.5	7	8	4	8
G162 50	5	7	9	11	7	10
G162 51	5.1	7	9	11	7	10
G162 55	5.5	8	10	11	7	10
G162 60	6	9	11	11	7	10
G162 65	6.5	10	12	11	7	12
G162 70	7	10	12	11	7	12
G162 75	7.5	11	13	11	7	12
G162 80	8	11	13	12	8	12
G162 85	8.5	13	15	12	8	15
G162 90	9	13	15	12	8	15
G162 95	9.5	14	16	12	8	15
G162 100	10	14	16	13	9	15
G162 110	11	15	17	13	9	15
G162 120	12	16	18	13	9	15
G162 130	13	17	19	13	9	15
G162 140	14	18	20	13	9	15
G162 150	15	19	21	13	9	15
G162 160	16	20	22	13	9	15
G162 170	17	21	23	13	9	15
G162 180	18	22	24	13	9	15
G162 190	19	23	25	13	9	15
G162 200	20	25	27	14	10	15

"G162" is used for more practical mounting of Bottle-neck Punches on the retainer plate.
Advantages:
* Cylindrical counter bore in the retainer plate is sufficient.
* Tempered Mounting Ring reduces the shrinking in the untempered retainer plate.
* You can take advantage of the positive features of Bottle-neck Punches without complex installation.

Material: Tempered Steel



Bottle-neck Punch (Blank type)

Code: **UBZ**

DIN 5118 (Trambone type)

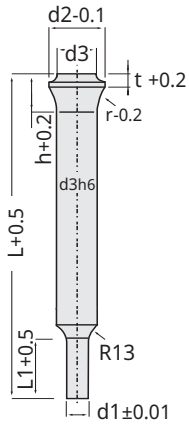
The TECINMAK UBZ Series is precisely engineered for high-demand applications where strong retraction force is essential.

Order Code	d1	L	d2	h	r
UBZ020080	2	80	3	3	3.5
UBZ020100		100			
UBZ021100	2.1	100	3.2	3	5
UBZ022100	2.2	100	3.2	3	5
UBZ023100	2.3	100	3.5	3	5
UBZ024100	2.4	100	3.5	3	5
UBZ025080	2.5	80	3.5	3	5
UBZ025100		100			
UBZ026100	2.6	100	4	3	6.5
UBZ027100	2.7	100	4	3	6.5
UBZ028100	2.8	100	4	3	6.5
UBZ029100	2.9	100	4	3	6.5
UBZ030080	3	80	4.5	3	6.5
UBZ030100		100			
UBZ031100	3.1	100	4.5	3	6.5
UBZ032100	3.2	100	4.5	3	6.5
UBZ033100	3.3	100	4.5	3	6.5
UBZ034100	3.4	100	4.5	3	6.5
UBZ035080	3.5	80	5	3	8
UBZ035100		100			
UBZ036100	3.6	100	5	3	8
UBZ037100	3.7	100	5	3	8
UBZ038100	3.8	100	5	3	8
UBZ039100	3.9	100	5	3	8
UBZ040080	4	80	5.5	4	8
UBZ040100		100			
UBZ041100	4.1	100	5.5	4	8
UBZ042100	4.2	100	5.5	4	8
UBZ043100	4.3	100	5.5	4	8
UBZ044100	4.4	100	5.5	4	8
UBZ045080	4.5	80	6	4	8
UBZ045100		100			
UBZ046100	4.6	100	6	4	8
UBZ047100	4.7	100	6	4	8
UBZ048100	4.8	100	6	4	8
UBZ049100	4.9	100	6	4	8
UBZ050080	5	80	7	4	10
UBZ050100		100			
UBZ051100	5.1	100	7	4	10
UBZ052100	5.2	100	7	4	10
UBZ053100	5.3	100	7	4	10
UBZ054100	5.4	100	7	4	10
UBZ055080	5.5	80	8	4	10
UBZ055100		100			
UBZ056100	5.6	100	8	4	10
UBZ057100	5.7	100	8	4	10
UBZ058100	5.8	100	8	4	10
UBZ059100	5.9	100	8	4	10
UBZ060080	6	80	9	4	10
UBZ060100		100			
UBZ061100	6.1	100	9	4	10
UBZ062100	6.2	100	9	4	10
UBZ063100	6.3	100	9	4	10
UBZ064100	6.4	100	9	4	10
UBZ065100	6.5	100	10	4	12
UBZ070100	7	100	10	4	12
UBZ075100	7.5	100	11	4	12
UBZ080080	8	80	11	4	12
UBZ080100		100			
UBZ085100	8.5	100	13	4	15
UBZ090100	9	100	13	4	15
UBZ095100	9.5	100	14	4	15
UBZ100100	10	100	14	4	15
UBZ105100	10.5	100	15	4	15
UBZ110100	11	100	15	4	15
UBZ115100	11.5	100	16	4	15
UBZ120100	12	100	16	4	15
UBZ125100	12.5	100	17	4	15
UBZ130100	13	100	17	4	15
UBZ135100	13.5	100	18	4	15
UBZ140100	14	100	18	4	15
UBZ150100	15	100	19	4	15
UBZ160100	16	100	20	4	15
UBZ170100	17	100	21	4	15
UBZ180100	18	100	22	4	15
UBZ190100	19	100	23	4	15
UBZ200100	20	100	25	4	15

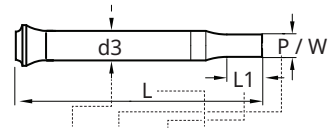
Material: 1.3343 (M2) **Hardness:** 62 - 66 HRC
Head Hardness: 45 - 55 HRC

NEW





How to order:
Shape
d3
P/W
L1
L
Material



UBZY x 10 x P7.0 x 13 x 80 - M2
UBZO x 10 x P8.2 x W5.2 x 25 x 100 - M2

Bottle-neck Stepped Punch DIN 5118 (Trambone type)

Code: **UBZ..**

The TECINMAK UBZ Series is precisely engineered for high-demand applications where strong retraction force is essential. Apart from our standards, special dimensions and complex geometries can be produced according to your technical drawings.

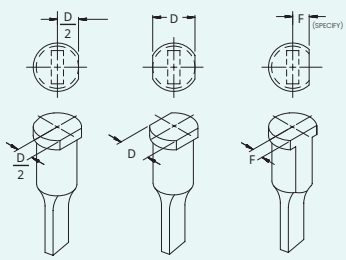
Please refer to the table on the right side

UBZY - UBZO - UBZK - UBZD - UBZA - UBZM - UBZX - UBZP

Order	d3	Head d2	t	h	r	d1 / Shape (UBZY) Round P	d1 / Shape Other Shapes W G/P	Standard L1	Alternative L1 Min. Max.	L
UBZ.. 03	3	4.5	3	6.03	6.5	1.00 ~ 2.95	1.00 ~ 2.95	10	13 - 19	50 63 71 80 90 100
UBZ.. 04	4	5.5	4	7.38	8	1.50 ~ 3.95	1.50 ~ 3.95	13	10 - 19	
UBZ.. 05	5	7	4	8.36	10	1.50 ~ 4.95	1.50 ~ 4.95	13	10 - 19	
UBZ.. 06	6	9	4	9.27	10	1.60 ~ 5.95	1.60 ~ 5.95	13	10 - 19	
UBZ.. 08	8	11	4	9.81	12	2.50 ~ 7.95	2.00 ~ 7.95	19	13 - 25	
UBZ.. 10	10	14	4	11.48	15	3.20 ~ 9.95	3.50 ~ 9.95	19	13 - 25	
UBZ.. 13	13	17	4	11.48	15	5.00 ~ 12.95	4.50 ~ 12.95	19	13 - 25	
UBZ.. 16	16	20	4	11.48	15	8.00 ~ 15.95	6.50 ~ 15.95	19	13 - 25	
UBZ.. 20	20	25	4	12.29	15	10.00 ~ 19.95	8.00 ~ 19.95	19	13 - 25	71 - 80 90 - 100

FLATS ON HEADED PRODUCTS

K1 KEY FLAT | **K5** DOUBLE FLAT | **K6** BODY FLAT



K1 and K5 are ground flush to shank. **K6** is a user defined flat and requires an **"F"** dimension specified with order.
K5 flats are standard 180° apart.

Order Example with Flat:
UBZO x 10 x P8.2 x W5.2 x 25 x 100 - **K1** - M2
Keying Info

Order Example with Flat and Coating:
UBZO x 10 x P8.2 x W5.2 x 25 x 100 - K1 - M2 - **Duplex**
Coating Info

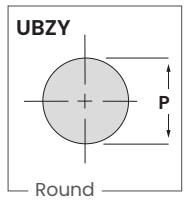
OPTIONAL FEATURES

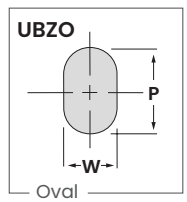
COATING

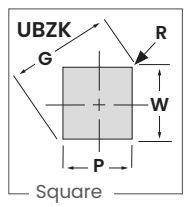


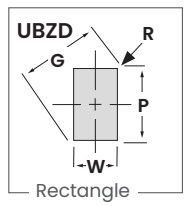
For details on Keying Information / Key Flat Options, More Complex Special Shapes, Shear Angles, and Coatings, please refer to the beginning of Category 6.

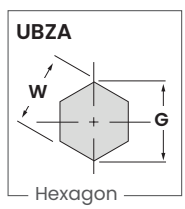
SHAPES

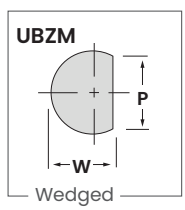


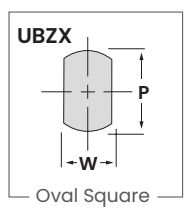


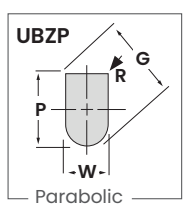


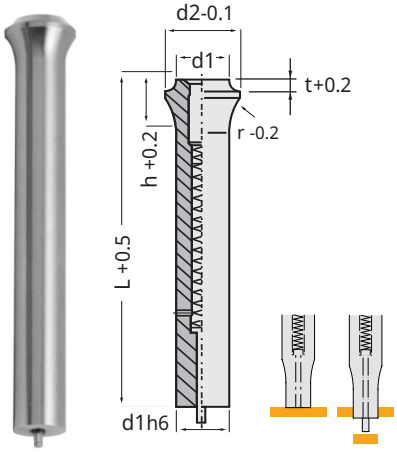












Code: **UFZ** (Blank)

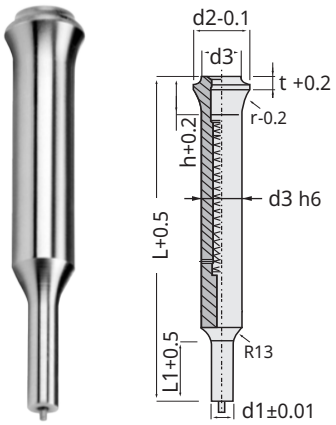
Bottle-neck Punch with Ejector

DIN 5118 (Trambone type)

Order Code	d1	L	d2	t	h	r
UFZ050080	5	80	7	4	8.36	10
UFZ050100		100				
UFZ060080	6	80	9	4	9.27	10
UFZ060100		100				
UFZ070080	7	80	10	4	9.81	12
UFZ070100		100				
UFZ080080	8	80	11	4	9.81	12
UFZ080100		100				
UFZ090100	9	100	13	4	11.48	15
UFZ100100	10	100	14	4	11.48	15
UFZ105100	10.5	100	14.5	4	11.48	15
UFZ110100	11	100	15	4	11.48	15
UFZ120100	12	100	16	4	11.48	15
UFZ130100	13	100	17	4	11.48	15
UFZ150100	15	100	19	4	11.48	15
UFZ155100	15.5	100	19.5	4	11.48	15
UFZ160100	16	100	20	4	11.48	15
UFZ200100	20	100	25	4	12.29	15
UFZ250100	25	100	31	4	13	15

Material: 1.3343 (M2)
Hardness: 62 - 66 HRC
Head Hardness: 45 - 55 HRC

NEW



Code: **UFZ..**

Bottle-neck Stepped Punch with Ejector

DIN 5118 (Trambone type)

The TECINMAK UFZ Series is precisely engineered for high-demand applications where strong retraction force is essential. Apart from our standards, special dimensions and complex geometries can be produced according to your technical drawings.

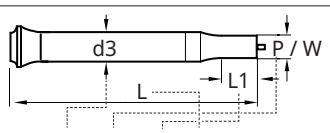
Please refer to the table on the right side

UFZY - UFZO - UFZK - UFZD - UFZA - UFZM - UFZX - UFZP

Order	d3	d1 / Shape (UFZY) Round P	d1 / Shape Other Shapes W G/P	L1 Standard	L
UFZ.. 05	5	2.00 ~ 4.95	2.00 ~ 4.95	13	80 100
UFZ.. 06	6	2.00 ~ 5.95	2.00 ~ 5.95	13	
UFZ.. 08	8	4.00 ~ 7.95	3.00 ~ 7.95	19	
UFZ.. 10	10	5.00 ~ 9.95	3.50 ~ 9.95	19	
UFZ.. 13	13	6.00 ~ 12.95	4.50 ~ 12.95	19	
UFZ.. 16	16	8.00 ~ 15.95	6.50 ~ 15.95	19	
UFZ.. 20	20	10.00 ~ 19.95	8.00 ~ 19.95	19	
UFZ.. 25	25	12.00 ~ 24.95	9.00 ~ 24.95	19	

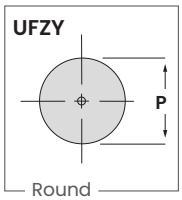
Order	Alternative L1 Min. Max.	Head d2	t	h	r
UFZ.. 05	10 - 19	7	4	8.36	10
UFZ.. 06	10 - 19	9	4	9.27	10
UFZ.. 08	13 - 25	11	4	9.81	12
UFZ.. 10	13 - 25	14	4	11.48	15
UFZ.. 13	13 - 25	17	4	11.48	15
UFZ.. 16	13 - 25	20	4	11.48	15
UFZ.. 20	13 ~ 25	25	4	12.29	15
UFZ.. 25	13 ~ 25	31	4	13	15

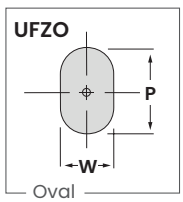
How to order:
Shape
d2
P/W
L1
L
Material

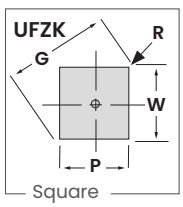


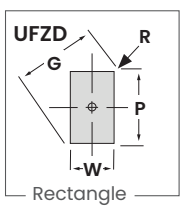
UFZY x 10 x P7.0 x 13 x 80 - M2
UFZO x 10 x P8.2 x W5.2 x 25 x 100 - M2

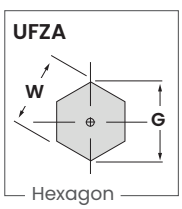
SHAPES

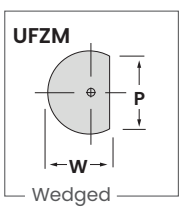


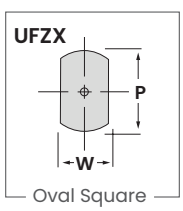


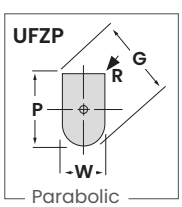


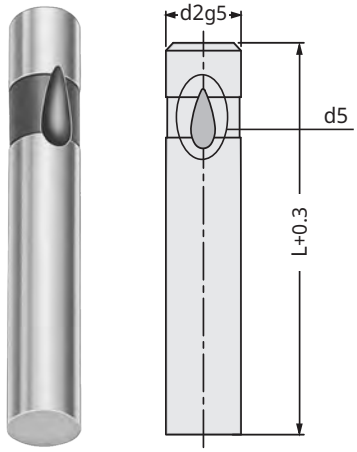












Code: **BHZ** Light Duty

Ball Lock Punch (Blank type)

ISO 10071

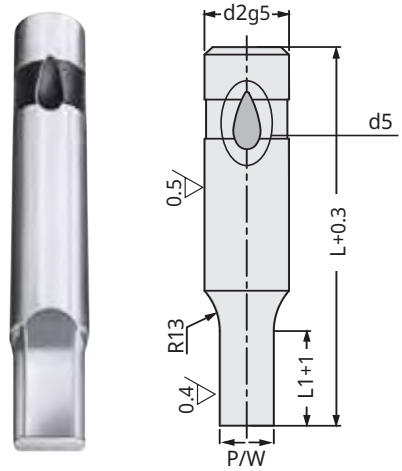
d2	d5	L
6	6	63
		71
		80
		90
		100
		125
10	8	63
		71
		80
		90
		100
		125
13	8	63
		71
		80
		90
		100
		125
16	8	63
		71
		80
		90
		100
		125

d2	d5	L
20	8	63
		71
		80
		90
		100
		125
25	8	63
		71
		80
		90
		100
		125
32	8	71
		80
		90
		100
		125
38	8	80
		90
		100
		125

Order: **BHZ.** d2 x L

Material: 1.3343 (M2)
Hardness: 60-62 HRC

Body and cutting edge are precision ground. Also, full or partial coating can be preferred upon request, by providing resistance against heat and friction on external layer, it solves problems such as winding and cold welding.



Code: **BH...** Light Duty

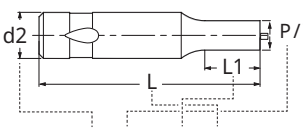
Ball Lock Stepped Punch

Please refer to the table on the right side

BHY - BHO - BHK - BHD - BHA -BHM - BHX - BHP

Order d2	Ball Socket d5	Shape (BHY) Round P	Shape Other Shapes W G/P	Standard L1	Alternative L1 Min. Max.	L
BH.. 06	6	2.2 ~ 5.9	2.2 ~ 5.9	13	10 -	63
BH.. 10	8	2.5 ~ 9.9	2.5 ~ 99	19	10 19	71
BH.. 13	8	5.0 ~ 12.9	4.5~12.9	19	13 25	80
BH.. 16	8	8.0 ~ 15.9	6.0~15.9	19	13 25	90
BH.. 20	8	12 ~ 19.9	8 ~19.9	19	13 25	100
BH.. 25	8	16 ~ 24.9	10~24.9	19	13 25	125
BH.. 32	8	24 ~ 31.9	12.5~31.9	25	25 30	80 90
BH.. 38	8	30 ~ 37.9	14~37.9	25	25 30	100 125

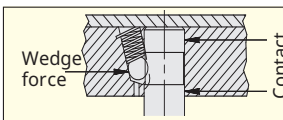
How to order:
Shape
d2
P/W
L1
L
Material



BHY x 10 x P8.2 x 19 x 100 - M2
BHO x 13 x P8.2 x W5.2 x 25 x 100 - M2

- When P = d2 shank / body tolerances apply.

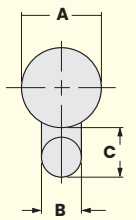
Material: 1.3343 (M2)
Hardness: 60- 62 HRC



Space Requir ements of Ball Locking Place

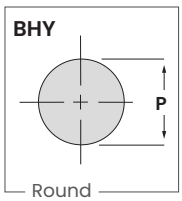
Punch Dia. A	Ball Dia. B	Clearance C
6	8	11
10		
13		
16		
20		
25		
32		
38		

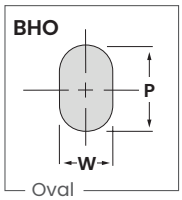
Connecting Punch to Retainer with Ball Socket

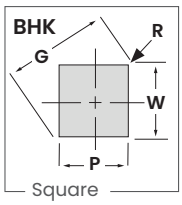


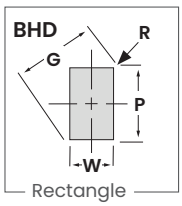
for precise retainers, please add 1.7 to **"C"** dimension

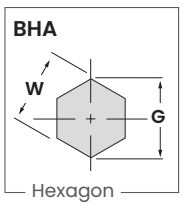
SHAPES

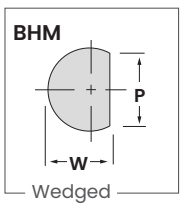


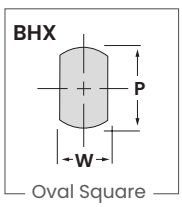


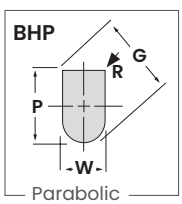


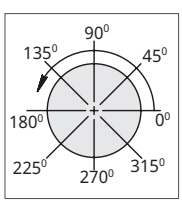


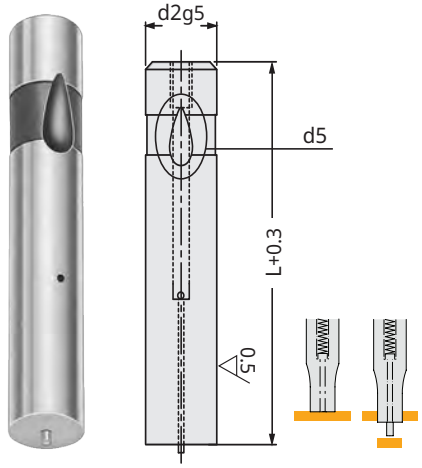












Code: **BFZ** Light Duty

Ball Lock Punch with Ejector

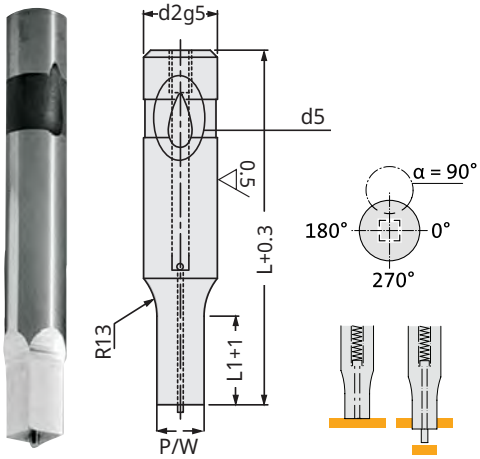
ISO 10071 (Blank type)

d2	d5	L
6	6	63
		71
		80
		90
		100
		125
10	8	63
		71
		80
		90
		100
		125
13	8	63
		71
		80
		90
		100
		125
16	8	63
		71
		80
		90
		100
		125
20	8	63
		71
		80
		90
		100
		125
25	8	63
		71
		80
		90
		100
		125
32	8	71
		80
		90
		100
		125
38	8	80
		90
		100
		125

Order: **BFZ.** d2 x L

Material: 1.3343 (M2)
Hardness: 60-62 HRC

Body and cutting edge are precision ground.
Also, full or partial coating can be preferred upon request, by providing resistance against heat and friction on external layer, it solves problems such as winding and cold welding.



Code: **BF...** Light Duty

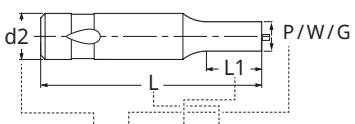
Ball Lock Stepped Punch with Ejector

Please refer to the table on the right side

BFY - BFO - BFK - BFD - BFA -BFM - BFX - BFP

Order d2	Ball Socket d5	Shape (BFY) Round P	Shape Other Shapes W G/P	Standard L1	Alternative L1 Min. Max.	L
BF.. 06	6	2.2 ~ 5.9	2.2 ~ 5.9	13	10 -	63 71 80 90 100 125
BF.. 10	8	2.5 ~ 9.9	2.5 ~ 99	19	10 19	
BF.. 13	8	5.0 ~ 12.9	4.5~12.9	19	13 25	
BF.. 16	8	8.0 ~ 15.9	6.0~15.9	19	13 25	
BF.. 20	8	12 ~ 19.9	8 ~19.9	19	13 25	
BF.. 25	8	16 ~ 24.9	10~24.9	19	13 25	
BF.. 32	8	24 ~ 31.9	12.5~31.9	25	25 30	80 90 100 125
BF.. 38	8	30 ~ 37.9	14~37.9	25	25 30	

How to order:
Shape
d2
P/W
L1
L
Material



BFY x 10 x P8.2 x 19 x 100 - M2
BFO x 13 x P8.2 x W5.2 x 25 x 100 - M2

- When P = d2 shank / body tolerances apply.

Material: 1.3343 (M2)
Hardness: 60- 62 HRC

Round / **P** +01 / -00
.01 From **P** to **d2**

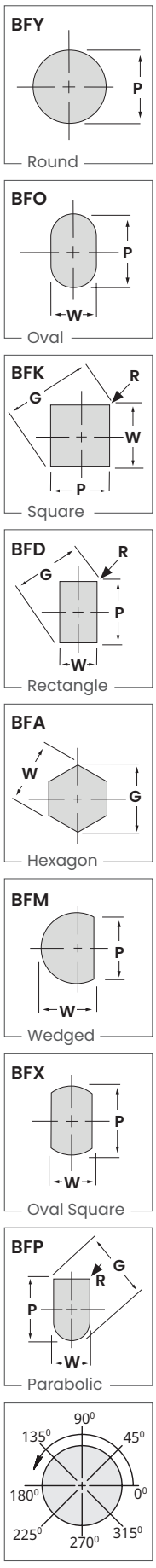
Shape / **P/W/G** ±01
.02 From **P** to **d2**

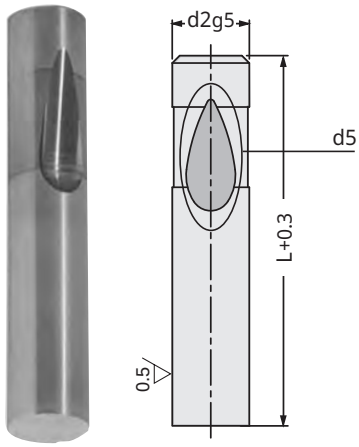
Standard Tolerances

Standard position of ball socket is 90°.
It can be preferred as 0°- 180°- 270°.

Note: Parts are viewed in die position looking from above the die. Punches are viewed looking through the body. Matrixes are viewed through top face.

SHAPES





Code: **BAZ** Heavy Duty

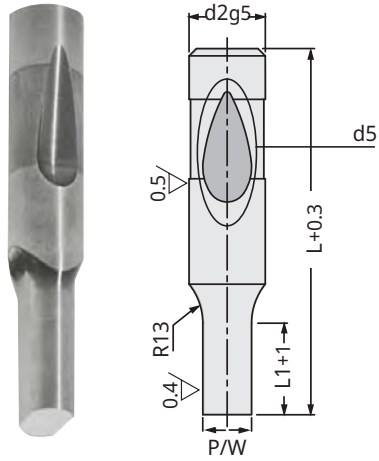
Ball Lock Punch (Blank type)

d2	d5	L
10	10	71
		80
		90
		100
13	12	71
		80
		90
		100
		125
16	12	71
		80
		90
		100
		125
20	12	71
		80
		90
		100
		125
25	12	80
		90
		100
		110
		125
32	12	80
		90
		100
		110
		125
40	12	80
		90
		100
		110
		125

Order: **BAZ.** d2 x L

Material: 1.3343 (M2)
Hardness: 60-62 HRC

Body and cutting edge are precision ground. Also, full or partial coating can be preferred upon request, by providing resistance against heat and friction on external layer, it solves problems such as winding and cold welding.



Code: **BA...** Heavy Duty

Ball Lock Stepped Punch

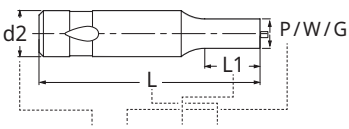
Please refer to the table on the right side

BAY - BAO - BAK - BAD - BAA -BAM - BAX - BAP

Order d2	Ball Socket d5	Shape (BAY) Round P	Shape Other Shapes W G/P	Standard L1	Alternative L1 Min. Max.	L
BA.. 10	10	2.5 ~ 9.8	2.5 ~ 10	19	10 19	71 80 90 100 110 125
BA.. 13	12	5 ~ 12.8	4.5 ~ 13	19	13 25	
BA.. 16	12	8 ~ 15.8	6 ~ 16	19	13 25	
BA.. 20	12	12 ~ 19.8	8 ~ 20	19	13 25	
BA.. 25	12	16 ~ 24.8	10 ~ 25	19	13 25	
BA.. 32	12	24 ~ 31.8	12 ~ 32	19	13 25	
BA.. 40	12	30 ~ 39.8	14 ~ 40	25	19 30	

- BA.. Punches conform to NAAMS standards.

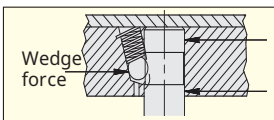
How to order:
Shape
d2
P/W
L1
L
Material



BAY x 10 x P5.5 x 19 x 90 - M2
BAO x 13 x P8.2 x W6.1 x 19 x 100 - M2

- When P = d2 shank / body tolerances apply.

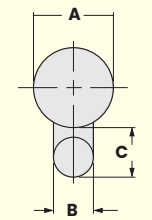
Material: 1.3343 (M2)
Hardness: 60- 62 HRC



Space Requirements of Ball Locking Place

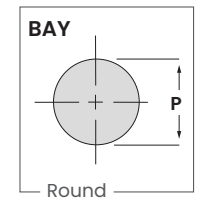
Punch Dia. A	Ball Dia. B	Clearance C
10	10	13
13	12	15
16		
20		
25		
32		
40		

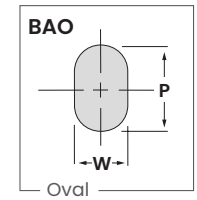
Connecting Punch to Retainer with Ball Socket

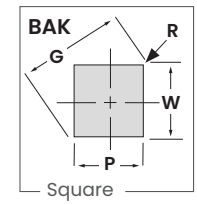


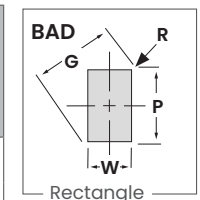
for precise retainers, please add 1.7 to **"C"** dimension

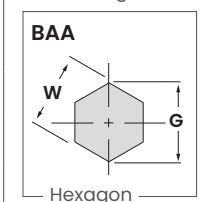
SHAPES

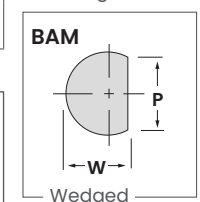


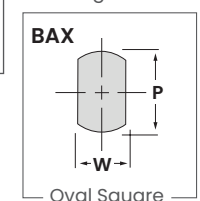


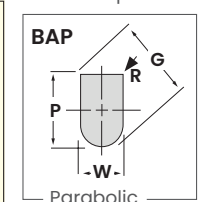


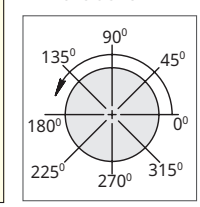


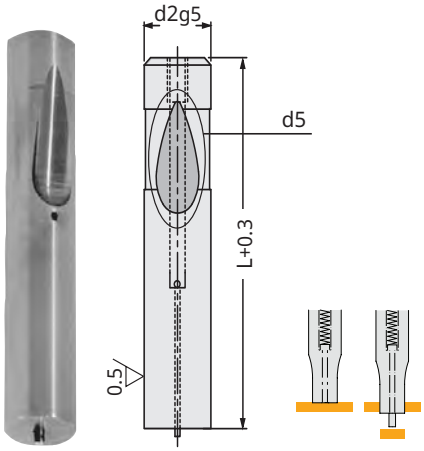












Code: **BAFZ** Heavy Duty

Ball Lock Punch with Ejector

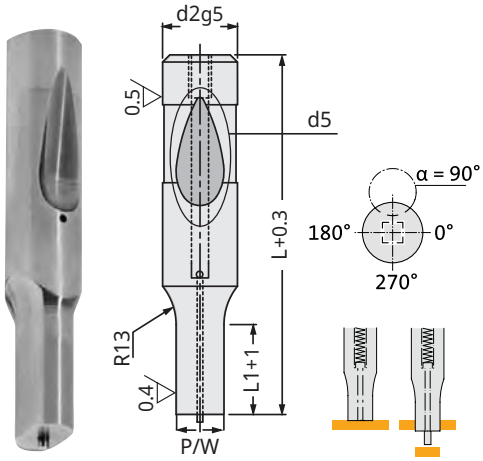
(Blank type)

d2	d5	L
10	10	71
		80
		90
		100
13	12	71
		80
		90
		100
		125
16	12	70
		80
		90
		100
		125
20	12	71
		80
		90
		100
		125
25	12	80
		90
		100
		110
		125
32	12	80
		90
		100
		110
		125
40	12	80
		90
		100
		110
		125

Order: **BAFZ.** d2 x L

Material: 1.3343 (M2)
Hardness: 60-62 HRC

Body and cutting edge are precision ground. Also, full or partial coating can be preferred upon request, by providing resistance against heat and friction on external layer, it solves problems such as winding and cold welding.



Code: **BAF...** Heavy Duty

Ball Lock Stepped Punch with Ejector

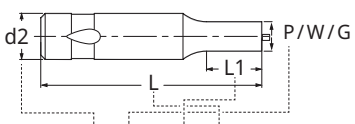
Please refer to the table on the right side

BAFY - BAFO - BAFK - BAFD - BAFA -BAFM - BAFX - BAFP

Order d2	Ball Socket d5	Shape (BAFY) Round P	Shape Other Shapes W G/P	Standard L1	Alternative L1 Min. Max.	L
BAF.. 10	10	2.5 ~ 9.8	2.5 - 10	19	10 19	71 80 90 100 110 125
BAF.. 13	12	5.0 ~ 12.8	4.5 - 13	19	13 25	
BAF.. 16	12	8.0 ~ 15.8	6.0 - 16	19	13 25	
BAF.. 20	12	12 ~ 19.8	8.0 - 20	19	13 25	
BAF.. 25	12	16 ~ 24.8	10 - 25	19	13 25	
BAF.. 32	12	24 ~ 31.8	12 - 32	19	13 25	
BAF.. 40	12	30 ~ 39.8	14 - 40	25	19 30	

- BAF.. Punches conform to NAAMS standards.

How to order:
Shape
d2
P/W
L1
L
Material



BAFY x 10 x P8.2 x 19 x 100 - M2
BAFO x 13 x P8.2 x W5.2 x 25 x 100 - M2

- When P = d2 shank / body tolerances apply.

Material: 1.3343 (M2)
Hardness: 60- 62 HRC

Round / **P** +01 / -00
.01 From **P** to **d2**

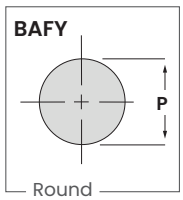
Shape / **P/W/G** ±01
.02 From **P** to **d2**

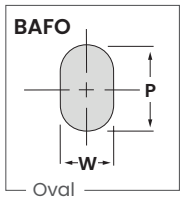
Standard Tolerances

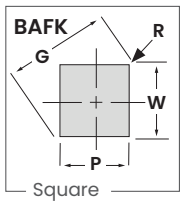
Standard position of ball socket is 90°.
It can be preferred as 0°- 180°- 270°.

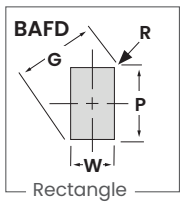
Note: Parts are viewed in die position looking from above the die. Punches are viewed looking through the body. Matrixes are viewed through top face.

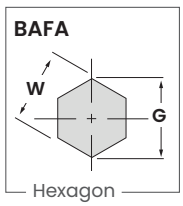
SHAPES

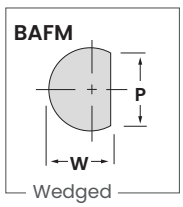


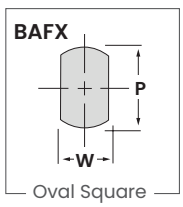


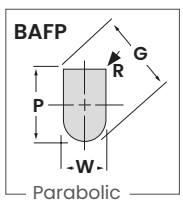


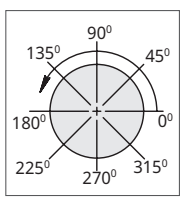


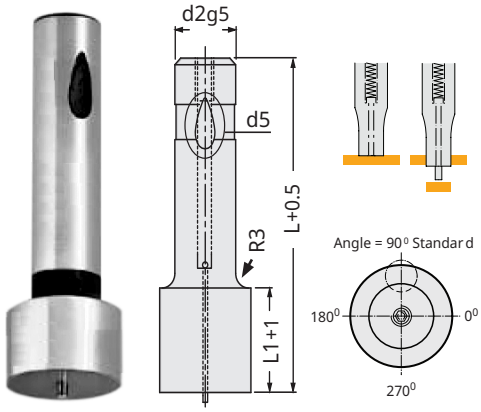












ROUND TYPE PUNCHES

Code: **BG..** Light Duty

Larger Ball Lock Ejector Punch

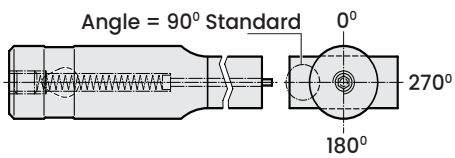
Body and outer dia. are precision ground.

Order d2	Ball Socket d5	P (round) Min. Max.	L1 Min. - Max.	L
BGY. 10	8	10.1 ~ 25	16	
BGY. 13	8	13.1 ~ 32	20	
BGY. 16	8	16.1 ~ 38	25	80
BGY. 20	8	20.1 ~ 40	25	90
BGY. 25	8	25.1 ~ 44	25	100
BGY. 32	8	32.1 ~ 50	32	
BGY. 40	8	40.1 ~ 63	32	

Round / Shaped — Order: **BG..** d2 x P/W/G x L1 x L

Please refer to the table on the right side
BGY - BGO - BGK - BGD - BGA - BGM - BGX - BGP

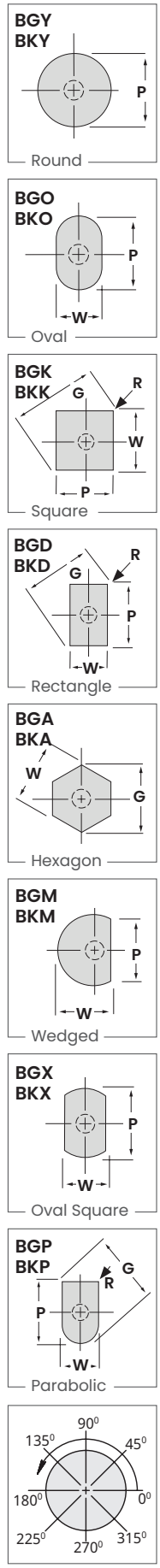
SHAPED TYPE PUNCHES

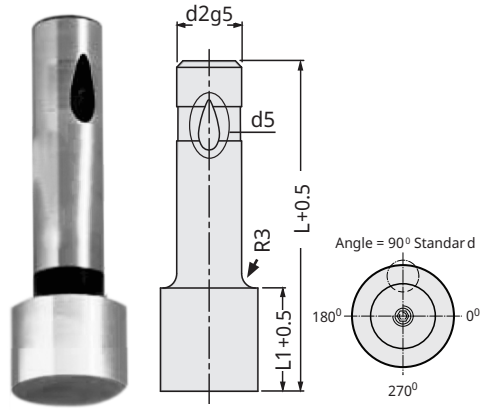


Order d2	Ball Socket d5	P/W/G (shape) Min. Max.	L1 Min. - Max.	L
BG.. 10	8	3.0 ~ 25	16	
BG.. 13	8	5.0 ~ 32	20	
BG.. 16	8	6.5 ~ 38	25	80
BG.. 20	8	8.0 ~ 40	25	90
BG.. 25	8	10 ~ 44	25	100
BG.. 32	8	11.5 ~ 50	32	
BG.. 40	8	14 ~ 63	32	

Note: Special dimensions on request.

SHAPES





ROUND TYPE PUNCHES

Code: **BK..** Light Duty

Larger Ball Lock Punch

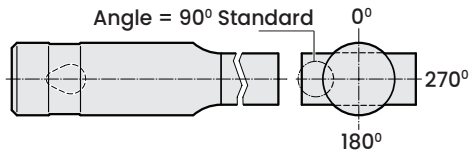
Body and outer dia. are precision ground.

Order d2	Ball Socket d5	P (round) Min. Max.	L1 Min. - Max.	L
BKY. 10	8	10.1 ~ 25	16	
BKY. 13	8	13.1 ~ 32	20	
BKY. 16	8	16.1 ~ 38	25	80
BKY. 20	8	20.1 ~ 40	25	90
BKY. 25	8	25.1 ~ 44	25	100
BKY. 32	8	32.1 ~ 50	32	
BKY. 40	8	40.1 ~ 63	32	

Round / Shaped — Order: **BK..** d2 x P/W/G x L1 x L

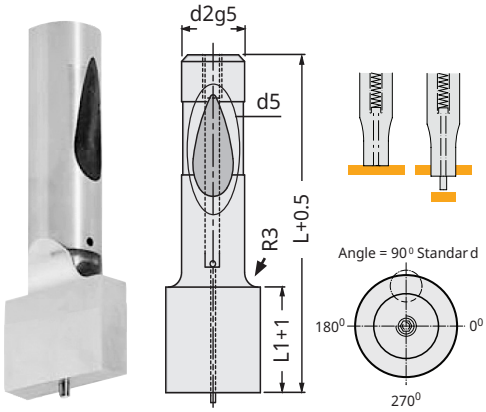
Please refer to the table on the left side
BKY - BKO - BKK - BKD - BKA - BKM - BKX - BKP

SHAPED TYPE PUNCHES

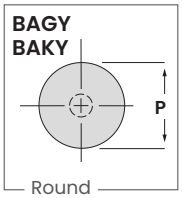


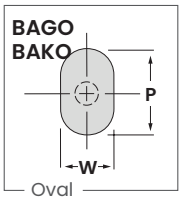
Order d2	Ball Socket d5	P/W/G (shape) Min. Max.	L1 Min. - Max.	L
BK.. 10	8	3.0 ~ 25	16	
BK.. 13	8	5.0 ~ 32	20	
BK.. 16	8	6.5 ~ 38	25	80
BK.. 20	8	8.0 ~ 40	25	90
BK.. 25	8	10 ~ 44	25	100
BK.. 32	8	11.5 ~ 50	32	
BK.. 40	8	14 ~ 63	32	

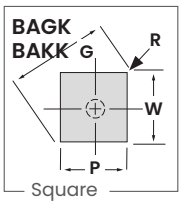
Note: Special dimensions on request.

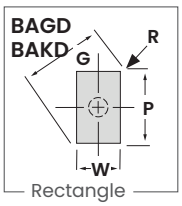


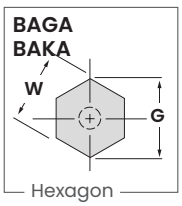
SHAPES

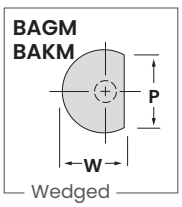


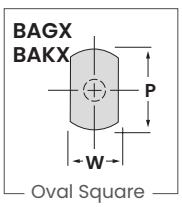


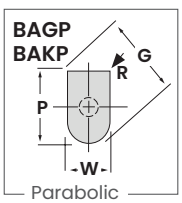


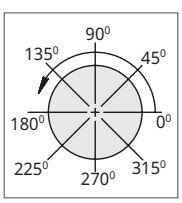












ROUND TYPE PUNCHES

Code: **BAG..** Heavy Duty

Larger Ball Lock Ejector Punch

Body and outer dia. are precision ground.

Order d2	Ball Socket d5	P (round) Min. Max.	L1 Min. - Max.	L
BAGY. 10	10	10.1 ~ 25	16	
BAGY. 13	12	13.1 ~ 32	20	
BAGY. 16	12	16.1 ~ 38	25	80
BAGY. 20	12	20.1 ~ 40	25	90
BAGY. 25	12	25.1 ~ 47	25	100
BAGY. 32	12	32.1 ~ 63	32	
BAGY. 40	12	40.1 ~ 63	32	

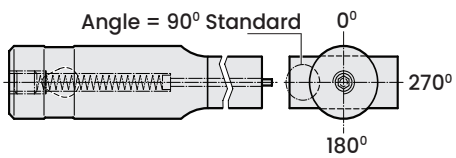
- BAG.. Punches conform to NAAMS standards.

Round Shaped — Order: **BAG..** d2 x P/W/G x L1 x L

Please refer to the table on the right side

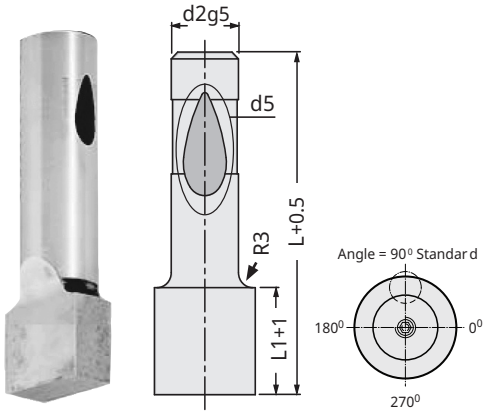
BAGY - BAGO - BAGK - BAGD - BAGA - BAGM - BAGX - BAGP

SHAPED TYPE PUNCHES



Order d2	Ball Socket d5	P/W/G (shape) Min. Max.	L1 Min. - Max.	L
BAG.. 10	10	3.0 ~ 25	16	
BAG.. 13	12	5.0 ~ 32	20	
BAG.. 16	12	6.5 ~ 38	25	80
BAG.. 20	12	8.0 ~ 40	25	90
BAG.. 25	12	10 ~ 47	25	100
BAG.. 32	12	11.5 ~ 63	32	
BAG.. 40	12	14 ~ 63	32	

Note: Special dimensions on request.



ROUND TYPE PUNCHES

Code: **BAK..** Heavy Duty

Larger Ball Lock Punch

Body and outer dia. are precision ground.

Order d2	Ball Socket d5	P (round) Min. Max.	L1 Min. - Max.	L
BAKY. 10	10	10.1 ~ 25	16	
BAKY. 13	12	13.1 ~ 32	20	
BAKY. 16	12	16.1 ~ 38	25	80
BAKY. 20	12	20.1 ~ 40	25	90
BAKY. 25	12	25.1 ~ 47	25	100
BAKY. 32	12	32.1 ~ 63	32	
BAKY. 40	12	40.1 ~ 63	32	

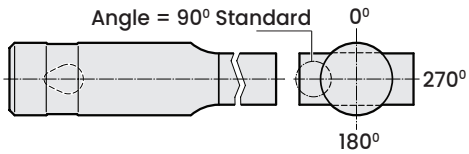
- BAK.. Punches conform to NAAMS standards.

Round Shaped — Order: **BAK..** d2 x P/W/G x L1 x L

Please refer to the table on the left side

BAKY - BAKO - BAKK - BAKD - BAKA - BAKM - BAKX - BAKP

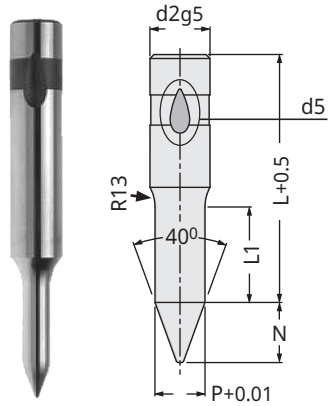
SHAPED TYPE PUNCHES



Order d2	Ball Socket d5	P/W/G (shape) Min. Max.	L1 Min. - Max.	L
BAK.. 10	10	3.0 ~ 25	16	
BAK.. 13	12	5.0 ~ 32	20	
BAK.. 16	12	6.5 ~ 38	25	80
BAK.. 20	12	8.0 ~ 40	25	90
BAK.. 25	12	10 ~ 47	25	100
BAK.. 32	12	11.5 ~ 63	32	
BAK.. 40	12	14 ~ 63	32	

Note: Special dimensions on request.

Pilots enter a previously punched hole to accurately position the workpiece and hold it in place before the main punches operate. This ensures precise and repeatable punching operations.



Code: **BPK** Light Duty

Ball Lock Pilot - **Conical**

d2	d5	P	L1	N	L
10	8	5.9 ~ 9.9	19	8	71
13	8	9.9 ~ 12.9	19	10	80
16	8	12.9 ~ 15.9	25	15	100
20	8	15.9 ~ 19.9	25	20	71
25	8	19.9 ~ 24.9	25	25	80
32	8	24.9 ~ 31.9	25	30	100
38	8	31.9 ~ 37.9	30	35	125
					140

Note: P / L selection upon request.
- Special dimensions on request.

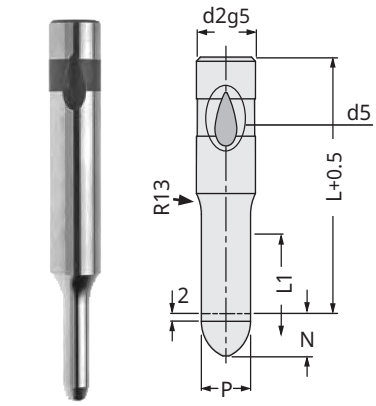
Order: **BPK.** d2 x P x L

Material: 1.3343 (M2)
Hardness: 60-62 HRC

"N" becomes = 1.2 x "P" minimum when "P" is below (see table).

Round / **P** +01 / -00 | .01 | From **P** to **d2**

- When P = d2 shank / body tolerances apply.



Code: **BPX** Light Duty

Ball Lock Pilot - **Spherical**

d2	d5	P	L1	N	L
6	6	2.9 ~ 5.9	13	-	71
10	8	5.9 ~ 9.9	19	8	80
13	8	9.9 ~ 12.9	19	10	100
16	8	12.9 ~ 15.9	25	15	71
20	8	15.9 ~ 19.9	25	15	80
25	8	19.9 ~ 24.9	25	20	100
32	8	24.9 ~ 31.9	25	25	125
					140

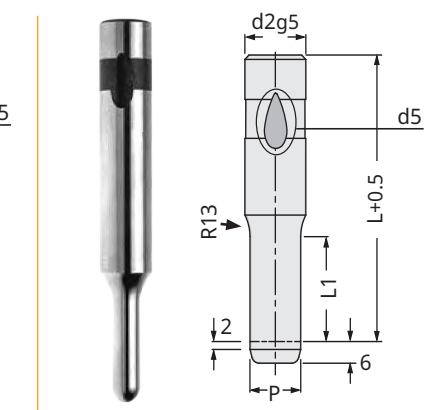
Note: P / L selection upon request.
- Special dimensions on request.

Order: **BPX.** d2 x P x L

Material: 1.3343 (M2)
Hardness: 60-62 HRC

- It is length of "L" pilot except end.
- 2 mm length is guided to punch before punch contacting sheet metal.

	P	N
10.1 mm	≤ 10 mm	8 mm
	- 15 mm	12 mm
	> 15 mm	15 mm



Code: **BPP** Light Duty

Ball Lock Pilot - **Parabolic**

d2	d5	P	L1	L
6	6	2.2 ~ 6.0	13	63
10	8	2.5 ~ 10	19	71
13	8	5 ~ 13	19	80
16	8	8 ~ 16	19	90
20	8	12 ~ 20	19	100
25	8	16 ~ 25	19	

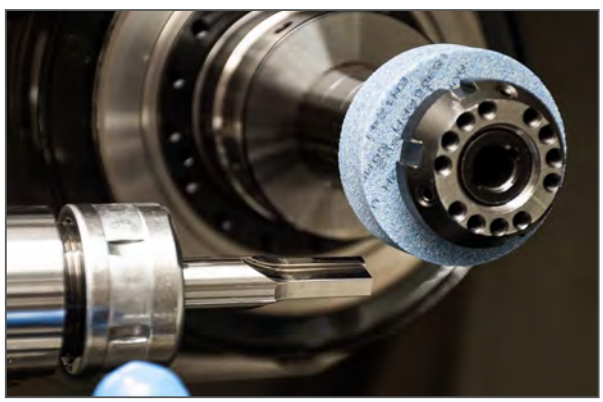
Note: P / L selection upon request.
- Special dimensions on request.

Order: **BPP.** d2 x P x L

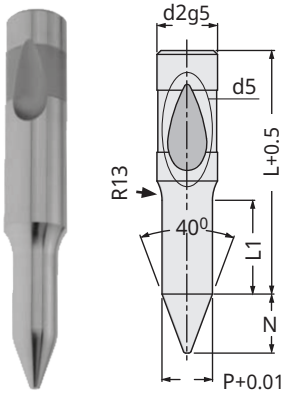
Material: 1.3343 (M2)
Hardness: 60-62 HRC

Round / **P** +01 / -00 | .01 | From **P** to **d2**

- When P = d2 shank / body tolerances apply.







Code: **BAPK** Heavy Duty

Ball Lock Pilot - **Conical**

d2	d5	P	L1	N	L
10	10	5.9 ~ 9.9	19	8	80
13	12	9.9 ~ 12.9	19	10	100
16	12	12.9 ~ 15.9	25	15	125
20	12	15.9 ~ 19.9	25	20	80
25	12	19.9 ~ 24.9	25	25	100
32	12	24.9 ~ 31.9	25	30	125
38	12	31.9 ~ 37.9	30	35	140
					150

Note: P / L selection upon request.
- Special dimensions on request.

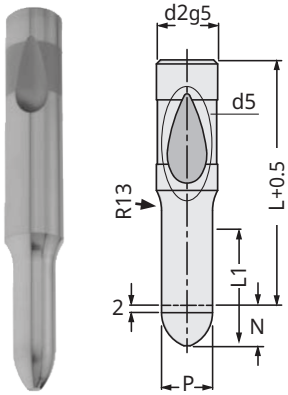
Order: **BAPK.** d2 x P x L

Material: 1.3343 (M2)
Hardness: 60-62 HRC

"N" becomes = 1.2 x "P" minimum when "P" is below (see table).

Round / **P** +01 / -00 | .01 | From **P** to **d2**

- When P = d2 shank / body tolerances apply.



Code: **BAPX** Heavy Duty

Ball Lock Pilot - **Spherical**

d2	d5	P	L1	N	L
10	10	2.5 ~ 10	19	10	71
13	12	5 ~ 13	19	10	80
16	12	8 ~ 16	19	10	100
20	12	12 ~ 20	19	10	71
25	12	16 ~ 25	19	10	80
32	12	24 ~ 32	19	10	100
40	12	30 ~ 40	25	10	125

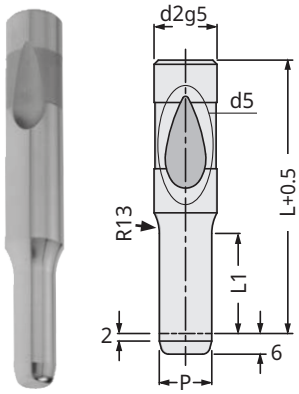
Note: P / L selection upon request.
- Special dimensions on request.

Order: **BAPX.** d2 x P x L

Material: 1.3343 (M2)
Hardness: 60-62 HRC

- It is length of "L" pilot except end.
- 2 mm length is guided to punch before punch contacting sheet metal.

	P	N
10.1 mm	≤ 10 mm	8 mm
	- 15 mm	12 mm
	> 15 mm	15 mm



Code: **BAPP** Heavy Duty

Ball Lock Pilot - **Parabolic**

d2	d5	P	L1	L
10	10	2.5 ~ 10	19	71
13	12	5 ~ 13	19	80
16	12	8 ~ 16	19	100
20	12	12 ~ 20	19	71
25	12	16 ~ 25	19	80
32	12	24 ~ 32	19	100
40	12	30 ~ 40	25	125

Note: P / L selection upon request.
- Special dimensions on request.

Order: **BAPP.** d2 x P x L

Material: 1.3343 (M2)
Hardness: 60-62 HRC

Round / **P** +01 / -00 | .01 | From **P** to **d2**

- When P = d2 shank / body tolerances apply.

"TECINMAK is the right address for high-quality Die & Mould Components, including precision punches and matrix, for all industrial applications."





